Supplementary Information

Microbial biogeography and functional differentiation across the Rub' al-Khali

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S1 Controls and contamination quantification

We followed recommended guidelines for reporting contamination in low biomass samples by analyzing each control according to the study stage it tested [16]. Table S1 details the controls included in the analyses. The canonical 1,271-profile amplicon table contains 24 sequenced assay-control profiles: 18 Trip 5 extraction blanks (`EB1`–`EB18`, of which 17 have confirmed extraction-day links to 217 biological profiles and `EB18` lacks an extraction-batch mapping) and six sequenced Trip 5 water PCR no-template controls (`Negative1`, `Negative2`, `Negative4`–`Negative7`). Six additional Trip 4 extraction blanks (`SEB`, `EB1`–`EB5`; 24 sequenced extraction blanks in total, 23 batch-linked across Trips 4 and 5) and four positive mock-community standards were denoised in separate control analyses, whereas reagent-only amplification-plate negative controls did not have a measurable DNA concentration and were not sequenced because TapeStation profiles showed only primer and adapter dimers. After excluding the 24 assay controls, ten low-depth biological profiles and duplicate sequencing runs were removed to yield the 1,186 deduplicated primary-transect profiles ([PENDING: `ECOL-READ-CUTOFF-RECONCILIATION`; current manuscript text cites a $< 10{,}000$ read cutoff whereas canonical data descriptors and companion methods record ten biological profiles with $< 1{,}000$ reads, including `T1Dr1` with 934 reads; source threshold under verification]).

S1.1 DNA extraction kit controls

Since the study involved DNA extraction using different kits and extractions during different sessions, we included extraction blanks to understand contamination introduced during the extraction steps. We performed two sensitivity analyses using extraction blanks linked to samples from two different campaigns (Trip 4 and Trip 5), independently. We denoised the six Trip 4 extraction blanks together with the additional control libraries and matched their ASVs to the biological count table by exact sequence. T5 samples and their extraction blanks shared a sequencing run. For both analyses, we considered an ASV as a candidate contaminant if it was present in at least 2 blanks and was more prevalent in blanks than in samples using an unadjusted one-sided Fisher test ($p < 0.1$). We quantified the candidate contaminant ASVs in the samples, then removed them and measured the effect of their removal on Shannon diversity and raw richness using Spearman's correlation coefficient to compare the sample profiles before and after their removal.

The Trip 5 set included 17 extraction blanks linked to 217 samples, and we found 351 candidate contaminant ASVs that comprised 2.2% of pooled reads and a median of 0.40% per sample (IQR 0.16—0.99%, maximum 56.60%). Shannon diversity and raw richness rankings were strongly correlated before and after removing the candidate contaminants ($\rho = 0.995$ and 0.999, respectively). Trip 4 had 6 extraction blanks linked to 95 samples, and we found 7 candidate contaminant ASVs, including *Nesterenkonia*, *Brevibacterium* and *Alcaligenes* and 4 ASVs without a genus assignment.

Table S1: Control samples overview across the canonical survey table and stage-specific control analyses.

Control	Stage and material	Sequenced and linked analysis counts
Survey extraction blanks (T5)	Extraction; nuclease-free water and extraction reagents	18 canonical profiles (`EB1`–`EB18`); 17 batch-linked to 217 T5 biological profiles
Survey PCR no-template controls (T5)	Amplification; PCR reagents and nuclease-free water	6 canonical profiles (`Negative1`, `Negative2`, `Negative4`–`Negative7`)
T4 extraction-blank rerun	Extraction; nuclease-free water and extraction reagents	6 separately denoised blanks linked to 95 T4 biological profiles (23 batch-linked blanks across T4–T5)
Reagent-only plate controls	Amplification; PCR reagents without water or DNA template	Not sequenced (only primer/adapter dimers)
T2 DNA community controls	Amplification and sequencing; ZymoBIOMICS HMW DNA Standard (D6322)	2 samples
T3 whole-cell community controls	Extraction, amplification and sequencing; ZymoBIOMICS Microbial Community Standard (D6300)	2 samples

Candidate contaminant ASVs were present in 33% of the samples and comprised 0.14% of pooled reads (maximum 14%). Across the 95 linked samples, before/after Shannon diversity and raw richness rankings were strongly correlated (Spearman $\rho = 0.999$ and 1.000, respectively).

S1.2 Microbial community standard controls

We included two different types of mock microbial community control samples in the study design. These controls included purified DNA ZymoBIOMICS HMW DNA Standard (D6322) in Trip 1 and Trip 2, and whole-cell ZymoBIOMICS Microbial Community Standard (D6300) which we later extracted in Trip 3. We used two community control replicates from Trip 2 and two from Trip 3 to estimate genus recovery based on the known community composition from the vendor. At the genus level, one Trip 2 community control and two Trip 3 controls recovered all the known genera present, with 31%, 100%, and 99.99% of their reads assigned to the expected genera, respectively.

S1.3 Well-to-well contamination

Use of 96-well plates for sequencing library preparation is known to introduce cross-contamination between physically close samples in microbiome studies [5]. We used 259 biological samples from T1 and their well positions to evaluate whether well-to-well contamination occurred in our study. Within each campaign and 96-well plate, we compared Bray–Curtis dissimilarities between pairs of samples from different sites. We defined adjacent wells as pairs of wells sharing an edge or a corner (Euclidean distance $\leq \sqrt{2}$), and we matched the pairs by geographic-distance decile and whether they came from the same or different soil compartments. For profiles listed in more than one well, we retained the first listed position. We calculated the mean difference in dissimilarity between adjacent and non-adjacent pairs, weighted by the number of adjacent pairs per stratum. The null distribution used 999 permutations of sample-to-well assignments within each plate (random seed

42). Adjacent wells were not more similar than matched non-adjacent wells. Dissimilarity difference was +0.005 (999 permutations, one-sided $p = 0.87$).

S1.4 Replicate variation

Replicate samples collected a few millimetres apart from the same compartment, site, and campaign were generally similar in diversity. The median difference in rarefied richness between a replicate and the mean of its replicate group was 11 % (Figure S1). In the remaining 42 groups, one replicate was usually poorer than the others (15 of 20 groups with three replicates). These discordant replicates had fewer sequencing reads than the other replicates in the same group (32 of 37 cases; median 114,000 versus 365,000 reads) and lacked genera that were common in the rest of the group (a median of 6 of the 50 most common genera missing, compared with 1). They were not distinguished by measured DNA concentration. These patterns indicate that most discordance reflects technical variation in low-yield samples, clustered in particular campaigns and sites, rather than differences between soils a few millimetres apart.

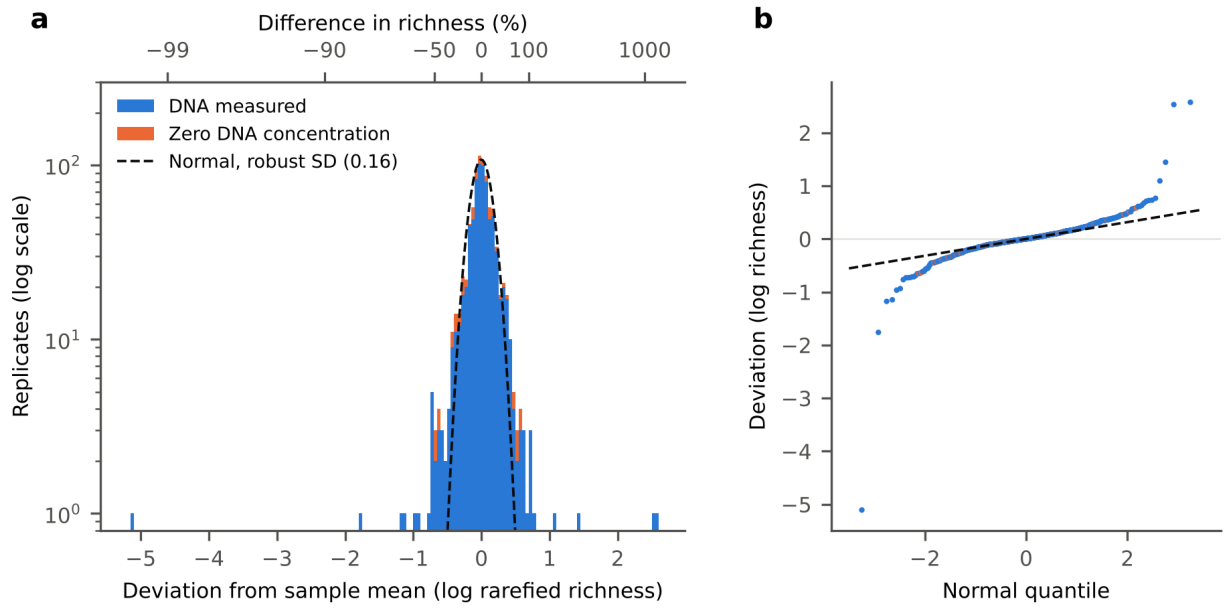


Figure S1: (a) Standard deviation differences in log richness between replicate samples. (b) Quantile–quantile (Q–Q) plot of the replicate deviations.

S2 XRF element axis

We retained the 11 elements detected in at least 75 % of laboratory XRF records: Al, Ca, Cl, Fe, K, Mg, Na, P, S, Si, and Ti. We transformed their readings as $\log(1 + x)$, and standardized each element within campaign. We then reduced the standardized elements to principal components and took the first component (PC1) as a single elemental score per sample (Table S2). PC1 captured 39.2 % of the elemental variance and was weighted most strongly toward Ca (0.413), Mg (0.406), Na (0.370), S (0.354) and Cl (0.338), and negatively toward Si (−0.273). This axis contrasts higher Ca, Mg, Na, S and Cl with lower Si. This score correlated $r = 0.825$ with an internal evaporite-element reference and $r = 0.976$ with a globally standardized version of the same axis.

We oriented PC1 to correlate positively with the mean within-campaign standardized log readings for Cl, Na and S, the retained elements in the internal reference set. The globally standardized sensitivity standardized each element across all campaigns together. Missing values and values constant within the standardization group were assigned zero after standardization.

Table S2: Elements used in the elemental score. Medians are taken over samples in which the element was detected. Loadings are the weights of each element on the first principal component (elemental score), computed on log-transformed concentrations standardized within campaign.

			Median by dune field				
Element	Detected (%)	Median (IQR)	Uruq	Seif	Sigm./hooked	Megabarchans	Loading
Ca	100.0	1.80 (0.89–3.93)	1.20	3.47	4.36	3.93	0.41
Mg	100.0	0.33 (0.24–0.51)	0.28	0.42	0.51	0.64	0.41
Na	96.8	0.21 (0.13–0.34)	0.18	0.26	0.26	0.50	0.37
S	87.6	0.052 (0.031–0.110)	0.042	0.056	0.077	0.480	0.35
Cl	87.3	0.069 (0.042–0.130)	0.052	0.066	0.092	0.970	0.34
Fe	100.0	0.52 (0.38–0.76)	0.49	0.60	0.58	0.52	0.28
Ti	98.6	0.12 (0.08–0.20)	0.11	0.14	0.14	0.12	0.28
K	99.9	0.78 (0.60–0.93)	0.74	0.82	0.86	0.75	0.17
Al	100.0	1.10 (0.86–1.40)	1.20	1.20	1.15	0.91	0.16
P	79.6	0.065 (0.041–0.120)	0.070	0.057	0.064	0.054	−0.01
Si	100.0	20.0 (16.9–21.8)	20.8	19.3	19.1	15.7	−0.27

S3 Relic DNA experiment

S3.1 Sample description

During T5, we collected additional samples from two campsites: C1R (root-adjacent soil) and C2R, C2S (root-adjacent soil, surface soil, respectively). We also collected C2 shallow sub-surface soil, but these samples were not included in the experiments because DNA extractions failed for these samples. For each of the three remaining parent soil tubes, representing a site × compartment group, we prepared 3 separate homogenized soil slurries in sterile phosphate-buffered saline (PBS, pH 7.4) at a 1 : 10 (w/v) ratio. We split each slurry into a treated and an untreated aliquot, giving nine paired comparisons grouped within three parent tubes.

S3.2 PMAxx treatment

We added PMAxx (Biotium, Hayward, USA) to each treated aliquot at a final concentration of 50 μM. We subjected both treated and untreated aliquots to the following conditions: 10 min dark incubation, photo activation with a 450 nm LED lamp for 35 min on a white reflective surface (15–20 cm from the lamp), and rotation at 15 min intervals. We stored samples at −20 °C until DNA extraction, which was performed using 860 μL from each aliquot with the DNeasy PowerSoil Pro kit (QIAGEN). We amplified the DNA and sequenced the samples with the protocols described in Section 0.1.

S3.3 Results

We calculated Hurlbert expected richness at 123,897 reads, the minimum depth among the 18 libraries, and Shannon diversity at full read depth. We report each treated–untreated difference

and the mean difference within each of the three site × compartment groups (Table S3; Figure S2). The headline-estimate index is under `analysis/v3/headline_uncertainty/`.

Table S3: Relic DNA aliquot comparisons for 9 sample pairs (18 libraries) across three site × compartment groups in T5 campsites.

Measure	C1R	C2R	C2S
Rarefied Richness (treated - untreated)	-36.70	-341.78	-111.49
Shannon (treated - untreated)	0.726	-0.025	-0.311

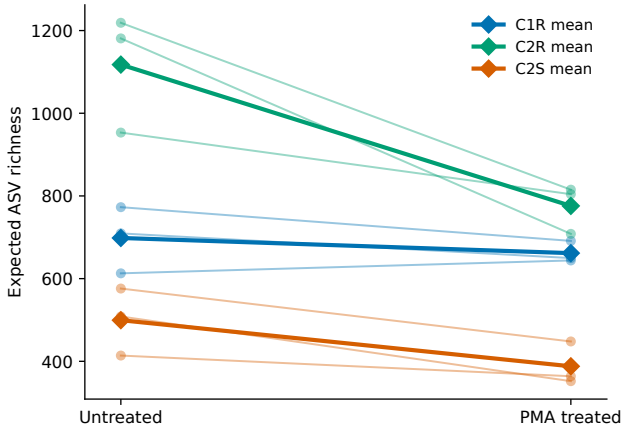


Figure S2: Expected ASV richness at 123,897 reads in untreated and PMA-treated aliquots. Thin lines join the paired aliquots from each slurry; thick lines show the mean of three slurries from each parent soil tube. C1R and C2R are root-adjacent soil at two campsites, and C2S is surface soil at the second campsite.

S4 Statistical Models

S4.1 Identifiability simulation

Environmental covariates and soil properties that change systematically across the landscape can be confounded with unmeasured spatial variation. We therefore used simulation to assess which associations our sampling design can identify. We report an association only when in simulation the 95% credible interval recovered the true value in at least 90% of replicates, and excluded zero in at most 10% of replicates in which there was no true effect.

We used the measured covariates and site coordinates, and simulated log richness for the 1,138 samples. Simulated values included compartment, campaign, and sequencing run effects, independent random effects for sites, visits (site × campaign) and units (site × campaign × compartment), and Student-t observation noise accounting for variation between replicate samples. All values were set to the posterior medians of the fitted richness model without covariates (standard deviations: site 0.08, visit 0.26, unit 0.33; residual scale 0.34). Spatial variation was added as a Matérn Gaussian field ($\nu = 1$) at the fitted range and standard deviation (SD) (967 km, SD = 0.56) or 200 km (SD = 0.4) generated independently of the covariates.

For site-mean elemental score, we simulated either no effect or an effect of -0.15 per SD, close to the observed estimate. For site mean pH, we simulated either no effect or an effect of -1.0 per

SD. Each replicate was fitted with the richness model described above, with an SPDE field using the same mesh and priors as the main analysis. We ran 100–300 replicates per scenario (Table S4). For each scenario we report coverage, the false-positive rate or power, bias, and their Monte Carlo standard errors.

We repeated the elemental-score checks for genera. We selected five genera spanning the range of prevalence (*Geodermatophilus*, *Chelativorans*, *Verrucosispora*, *Thermosipho* and *Staphylococcus*). For each genus, we simulated presence and abundance from its fitted hurdle model, with a true effect of either zero or -0.3 per SD, and ran 50 replicates per genus and scenario. This check used each genus's own coefficients. It did not cover the slopes expressed relative to the mean of all genera, or the false discovery rate correction.

Results. No fit failed or reached a degenerate solution in any scenario. With spatial variation but no covariate effect, the SPDE model rarely reported a false association for site-mean XRF (4.5% of replicates at the fitted range of 967 km and 3.5% at the 200 km range, against a nominal 5%). For site-mean pH, false associations were reported in 9% of replicates at the fitted range and 2% at the 200 km range. False-positive rates for within-site XRF were 7.0% (fitted range) and 12.0% (200 km), and for within-site pH 10.5% (fitted range) and 5.0% (200 km). The two highest rates were within about one Monte Carlo standard error of 10%. In these two scenarios, the model slightly underestimated uncertainty (mean posterior SD 0.021, against an SD of 0.024–0.025 in estimates across simulated datasets).

Estimates were unbiased in every simulated scenario with a covariate effect (absolute bias $\leq$ 0.005). A site-mean XRF effect of -0.15 per SD was detected in 92.5% of replicates (94.5% coverage), and a within-site XRF effect of -0.15 per SD in every replicate (96.0% coverage). Within-site pH, which had no effect, was falsely detected in 4.5% of these replicates. A site-mean pH effect of -0.10 per SD was detected in only 33% of replicates, although its intervals were well calibrated (91% coverage).

Genus models behaved similarly. With spatial variation but no covariate effect, false associations with site-mean XRF were reported in 8.0% of replicates for presence and 7.2% for abundance, and with within-site XRF and pH in 3.6–7.6%. Estimates were close to unbiased (absolute bias ≤ 0.018). A site-mean XRF effect of -0.3 per SD was detected in 51% (presence) and 72% (abundance) of replicates (coverage 92.8% and 89.6%), and a within-site XRF effect of -0.2 per SD in 49% and 78% (coverage 93.2% and 92.4%). The model slightly underestimated uncertainty for site-mean XRF: posterior SDs were about 12% smaller than the SD of estimates across simulated datasets.

Under random spatial variation, our sampling design can therefore identify associations with site-mean XRF and with within-site variation in XRF and pH, for both richness and genera (Table S4). Within-site associations with richness should be interpreted with some caution, as their false-positive rate reached 10–12% in one of the two spatial settings. Site-mean pH associations were well calibrated but had low power, so the absence of a detected pH association is not evidence of no effect. Power for genera was moderate, so a genus without a detected association may still respond to these properties. Monte Carlo standard errors were at most 4.7 percentage points (site-mean pH, 100 replicates) and at most 3.2 percentage points elsewhere.

S4.1.1 Prior sensitivity

We assessed whether the main estimates depended on the choice of priors. We refitted the compartment contrasts (models without environmental variables) and the association with site-mean XRF (models with the site mean and the spatial field). For each set, the median of the PC prior on the range of the spatial field was changed from 200 km to 50 km and to 500 km. For the

Table S4: Performance of the SPDE models in the identifiability simulation. Spatial variation was a Matérn Gaussian field at the range and SD of the fitted model (richness: 967 km, SD 0.56; genera: fitted per genus) or at a range of 200 km (SD 0.4; richness only). Richness scenarios had 200 replicates, except site-mean pH (100). Genus scenarios pooled five genera with 50 replicates each (250). "Excludes zero" is the percentage of replicates in which the 95% credible interval excluded zero: the false-positive rate when the true effect is zero, and the power otherwise. Coverage is the percentage in which the interval contained the true value (nominal 95%). Monte Carlo standard errors were ≤ 4.7 percentage points.

Response	Scenario	Term	True effect	Excludes zero (%)	Coverage (%)
Richness	No effect, fitted range	Site-mean XRF	0	4.5	95.5
		Site-mean pH	0	9.0	91.0
		Within-site XRF	0	7.0	93.0
		Within-site pH	0	10.5	89.5
	No effect, 200 km range	Site-mean XRF	0	3.5	96.5
		Site-mean pH	0	2.0	98.0
		Within-site XRF	0	12.0	88.0
		Within-site pH	0	5.0	95.0
	Site-mean XRF, fitted range	Site-mean XRF	−0.15	92.5	94.5
	Site-mean pH, fitted range	Site-mean pH	−0.10	33.0	91.0
	Within-site XRF, fitted range	Within-site XRF	−0.15	100.0	96.0
		Within-site pH	0	4.5	95.5
Genus presence	No effect, fitted range	Site-mean XRF	0	8.0	92.0
		Within-site XRF	0	6.4	93.6
		Within-site pH	0	5.6	94.4
	Site-mean XRF, fitted range	Site-mean XRF	−0.3	50.8	92.8
	Within-site XRF, fitted range	Within-site XRF	−0.2	48.8	93.2
		Within-site pH	0	6.4	93.6
Genus abundance	No effect, fitted range	Site-mean XRF	0	7.2	92.8
		Within-site XRF	0	3.6	96.4
		Within-site pH	0	7.6	92.4
	Site-mean XRF, fitted range	Site-mean XRF	−0.3	71.6	89.6
	Within-site XRF, fitted range	Within-site XRF	−0.2	77.6	92.4
		Within-site pH	0	6.4	93.6

compartment contrasts, we also changed the scale of the PC priors on the random-effect SDs from $P(\text{SD} > 1) = 0.01$ to $P(\text{SD} > 0.5) = 0.01$ and $P(\text{SD} > 2) = 0.01$, halving and doubling the prior scale. The prior on the SD of the spatial field was kept at $P(\sigma > 1) = 0.01$. All other model components were as in the main analysis. Prior sensitivity was not assessed for within-site associations, site-mean pH, climate or the genus models.

Results. Estimates were insensitive to the priors (Table S5). Across the four alternative priors, the compartment contrasts in rarefied richness changed by at most 0.1 percentage points, and those in Shannon and normalized Shannon by less than 0.001. Credible-interval limits changed by similar amounts, and no interval changed in whether it included zero. The site-mean XRF association with richness was −13.5% (95% CrI −21.0 to −5.2) with the 200 km prior, −13.5% (−20.9 to −5.3) with 50 km, and −13.4% (−20.9 to −5.3) with 500 km. The corresponding Shannon estimates ranged from −0.175 to −0.178 nats, and the normalized Shannon estimates from −0.015 to −0.018. All intervals excluded zero.

Table S5: Sensitivity of the main estimates to the priors. Richness contrasts are percentage changes; Shannon and normalized Shannon are differences. Main analysis: prior median range of the spatial field 200 km, and PC priors on the random-effect SDs with $P(\mathrm{SD} > 1) = 0.01$. Values are posterior medians with 95% credible intervals.

Estimate	Main analysis	Range 50 km	Range 500 km	SD scale ×0.5 / ×2
Rarefied richness (%)				
Shallow subsurface − surface	9.5 (0.5, 19.2)	9.5 (0.5, 19.2)	9.5 (0.5, 19.3)	9.5 (0.5, 19.2) / 9.5 (0.4, 19.3)
Root-adjacent − surface	−9.9 (−17.2, −1.8)	−9.9 (−17.1, −1.8)	−9.9 (−17.2, −1.8)	−9.9 (−17.2, −1.9) / −9.8 (−17.1, −1.7)
Root-adjacent − shallow subsurface	−17.7 (−24.3, −10.3)	−17.7 (−24.3, −10.3)	−17.7 (−24.3, −10.3)	−17.8 (−24.3, −10.4) / −17.7 (−24.3, −10.2)
Site-mean XRF (per SD)	−13.5 (−21.0, −5.2)	−13.5 (−20.9, −5.3)	−13.4 (−20.9, −5.3)	–
Shannon diversity (nats)				
Shallow subsurface − surface	0.11 (−0.01, 0.23)	0.11 (−0.01, 0.23)	0.11 (−0.01, 0.23)	0.11 (−0.01, 0.23) / 0.11 (−0.01, 0.23)
Root-adjacent − surface	−0.15 (−0.27, −0.03)	−0.15 (−0.27, −0.03)	−0.15 (−0.27, −0.03)	−0.15 (−0.27, −0.04) / −0.15 (−0.27, −0.03)
Root-adjacent − shallow subsurface	−0.26 (−0.38, −0.15)	−0.26 (−0.38, −0.15)	−0.26 (−0.38, −0.15)	−0.26 (−0.38, −0.15) / −0.26 (−0.38, −0.15)
Site-mean XRF (per SD)	−0.18 (−0.31, −0.04)	−0.18 (−0.32, −0.04)	−0.17 (−0.31, −0.04)	–
Normalized Shannon diversity				
Shallow subsurface − surface	0.012 (0.002, 0.023)	0.012 (0.002, 0.023)	0.012 (0.002, 0.023)	0.012 (0.002, 0.023) / 0.012 (0.002, 0.023)
Root-adjacent − surface	−0.029 (−0.039, −0.018)	−0.029 (−0.039, −0.018)	−0.029 (−0.039, −0.018)	−0.029 (−0.039, −0.018) / −0.029 (−0.039, −0.018)
Root-adjacent − shallow subsurface	−0.041 (−0.052, −0.031)	−0.041 (−0.052, −0.031)	−0.041 (−0.052, −0.031)	−0.041 (−0.052, −0.031) / −0.041 (−0.052, −0.031)
Site-mean XRF (per SD)	−0.016 (−0.028, −0.004)	−0.018 (−0.028, −0.006)	−0.015 (−0.027, −0.004)	–

S4.2 Models

Climate, soil properties and position change together along the transect, so we tested each type of association with the part of the data that can separate it from geography. Including a spatial field accounts for the similarity of nearby sites, but it does not by itself distinguish a measured variable from other factors that vary smoothly across the landscape [20]. We therefore first simulated log richness and genus presence and abundance on the real sampling design, with known environmental effects and with spatial trends unrelated to any measured variable, to test which associations the design could recover (Table S4).

We compared compartments sampled during the same site visit, in models without environmental variables. To test whether pH and XRF PC1 were associated with diversity within sites, we split each group's value into its site mean and its deviation from that mean [31], and entered both parts of both variables in one model. The deviations compare samples from the same site, so their associations are not confounded with differences between sites. Associations between sites were estimated in separate models, each including one site mean (pH or XRF PC1) and the spatial field, and therefore describe differences among sites beyond the broad spatial pattern. Long-term temperature and humidity closely track position along the transect ($r = 0.93$ and 0.92), so we modeled each separately, with and without the spatial field, and interpret their associations as descriptions of the west–east gradient rather than as independent environmental effects.

S4.3 Calibration of distance decay intervals

Pairwise distances share sites, so standard regression intervals for distance-decay slopes are too narrow. We used simulation to compare interval methods on our site layout. For each simulated dataset, we generated a CLR profile of 200 genera at each of the 60 sites. Each genus value was the sum of a site-level Gaussian process with exponential covariance and independent noise. Each genus had its own process. The range of the process was 300 or 1,000 km. We set the variances of the two terms so that the expected Aitchison distance matched the observed mean surface-soil distance for site pairs less than 50 km apart and more than 800 km apart. We then computed Aitchison distances between the simulated profiles and estimated the slope per 100 km by ordinary least squares, as in the main analysis. The true slope was the mean slope over 4,000 realizations of the same process.

We compared three interval methods: the delete-one-site jackknife; a delete-one-block jackknife, in which the sites were divided into 12, 6 or 4 contiguous blocks along the transect and each block was removed in turn (t quantiles with one fewer degree of freedom than the number of sites or blocks); and a model of the pairwise distances fitted with INLA, with random effects for both sites in each pair, either independent or spatially structured (BYM2 on a five-nearest-neighbor graph). We used 100 simulated datasets for the comparison with the INLA model and 500 for the comparison of jackknife intervals.

To assess intervals for the difference in slope between two compartments, we simulated pairs of compartments with the same true slope (300 datasets per range). Each compartment's profile combined a Gaussian process shared by both compartments, a compartment-specific Gaussian process with the same range, and independent noise. The shared fraction was chosen so that the correlation between the two compartments' distance matrices matched the observed value for shallow subsurface and root-adjacent soil (0.67). We compared the delete-one-site and 12-block jackknife intervals, and a permutation test of site labels (999 permutations) applied to the matrix of differences.

Results. Intervals that treated sites as exchangeable were too narrow (Table S6). Delete-one-site jackknife intervals contained the true slope in 82% of simulated datasets with a 300 km range and 67% with a 1,000 km range. Intervals from the INLA model of pairwise distances contained it in only 11–12%, whether site effects were independent or spatially structured. Block jackknife intervals reached or exceeded nominal coverage: 99.8% and 94% with 12 blocks, and 99–100% with 6 or 4 blocks. Block intervals were wider, 2.0–2.5 times the width of the site jackknife intervals with 12 blocks. We therefore used the 12-block jackknife, the smallest number of blocks that achieved nominal coverage at both ranges.

For differences between compartments, no method reached nominal coverage. Delete-one-site jackknife intervals contained the true difference of zero in 60–63% of datasets, and 12-block jackknife intervals in 77–81%. The site permutation test falsely detected a difference in 66–72% of datasets. Differences in decay slope between compartments should therefore be interpreted with caution.

Table S6: Coverage of 95% intervals for the distance-decay slope of Aitchison distance in simulated communities on the real site layout. Communities were generated from Gaussian processes with a range of 300 or 1,000 km. "Single compartment": percentage of simulated datasets in which the interval contained the true slope. "Difference": percentage in which the interval for the difference between two compartments with equal true slopes contained zero; for the permutation test, the percentage of datasets in which it falsely detected a difference ($p < 0.05$).

Target	Method	Range 300 km	Range 1,000 km
Single compartment	Delete-one-site jackknife[a]	82.2	66.8
	12-block jackknife[a]	99.8	94.0
	6-block jackknife[a]	100.0	99.0
	4-block jackknife[a]	100.0	100.0
	INLA pair model, independent sites[b]	12.0	11.0
	INLA pair model, BYM2 sites[b]	11.0	11.0
Difference	Delete-one-site jackknife[c]	63.3	60.0
	12-block jackknife[c]	80.7	76.7
	Site permutation test (false positives)[c]	66.0	71.7

[a]500 datasets; [b]100 datasets; [c]300 datasets.

S4.4 Genus models

S4.5 Hurdle models

We summed reads within each site × campaign × compartment group and retained groups with at least 2,000 reads (630 groups). Of the 384 genera present in at least 20% of these groups, we analyzed the 200 with the highest mean relative abundance. Most genera were absent from many samples, so for each genus we fitted a hurdle model with two parts, both at the level of individual samples (1,186 profiles). Presence was modeled with a binomial likelihood and standardized log total read count as a covariate. Abundance where present was modeled with a zero-truncated negative binomial likelihood, with the log total read count of the profile (including unclassified reads) as an offset. Both parts included the fixed effects of compartment, campaign and sequencing run, independent random effects for sites, visits and site × campaign × compartment groups, and the SPDE spatial field, with the same mesh and priors as the diversity models. Fixed effects for the design had normal priors with SD 10, because differences in genus abundance between compartments and campaigns can be large; standardized covariate slopes had normal priors with

SD 1.

Each genus was fitted with the same set of models as diversity (see Models). A model without environmental variables for compartment contrasts, a model with the site means and within-site deviations of pH and XRF PC1, and separate models with the site mean of XRF PC1 or of pH.

Read counts carry information only on relative abundance, so an increase in some genera lowers the share of all others. We therefore summarized each slope on two scales: the raw coefficient of each genus, and the coefficient relative to the mean coefficient of all 200 genera, computed for each of 20,000 posterior draws. The relative scale identifies genera that respond more or less strongly than the average genus and is unaffected by shifts shared by all genera, such as changes in total load or sequencing depth. For each slope we calculated a two-sided posterior tail probability and controlled the false discovery rate at 5% (Benjamini–Hochberg) separately for each model part, estimate and scale.

To compare dune fields, we added the four dune fields as a fixed effect (uruq as reference; normal prior with SD 10), alone or together with site-mean XRF. For each genus and model part, we tested whether the genus differed among dune fields with a Wald test of the three dune-field coefficients (3 degrees of freedom), and controlled the false discovery rate across genera.

S4.5.1 Joint model

We fitted hierarchical joint species distribution models (HMSC) [36] to the 50 most abundant of the 200 genera, with presence (probit) and log abundance where present (normal) modeled separately. Fixed effects were compartment, campaign, sequencing run and log read depth (HMSC does not use an offset). A second model added a natural spline of transect position (3 degrees of freedom) and the standardized site means of XRF PC1 and pH, so that associations that disappear in this model can be attributed to the shared west–east gradient. Random effects with latent factors were included at the site level (a Gaussian process on site coordinates), the site × campaign × compartment level and the sample level. We ran two chains with 12,500 burn-in iterations and retained 250 samples per chain at a thinning of 100.

Because most groups held only one or two samples, the group and sample levels exchanged covariance between chains, so we combined them into a single within-site level. For each posterior draw, we centered the within-site residual covariance on the mean of the 50 genera (centered log-ratio), which removes variation shared by all genera, and summarized each pair of genera by its proportionality ρ [29, 15]. A pair was classified as positively associated when $P(\rho > 0) > 0.95$ and negatively associated when $P(\rho > 0) < 0.05$. To test replication, we refitted the model separately to campaigns 1 and 3 and to campaigns 4 and 5. Convergence was assessed with the potential scale reduction factor.

S5 Figures

S6 Tables

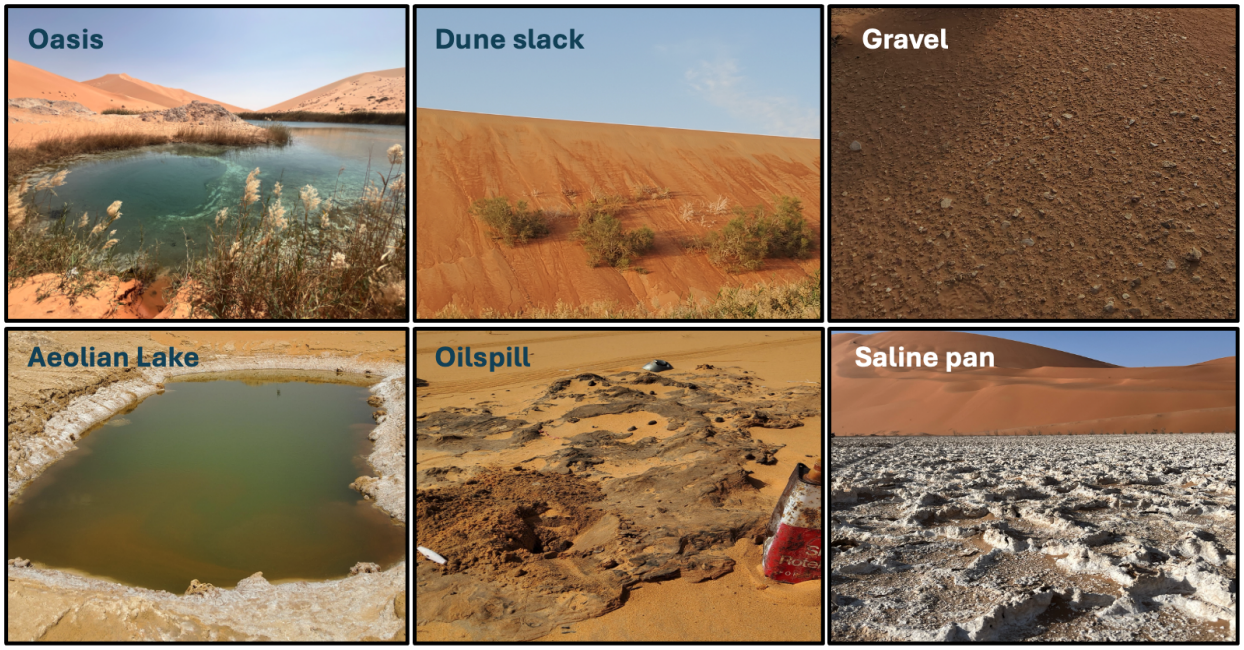


Figure S3: Landforms observed in the study area.

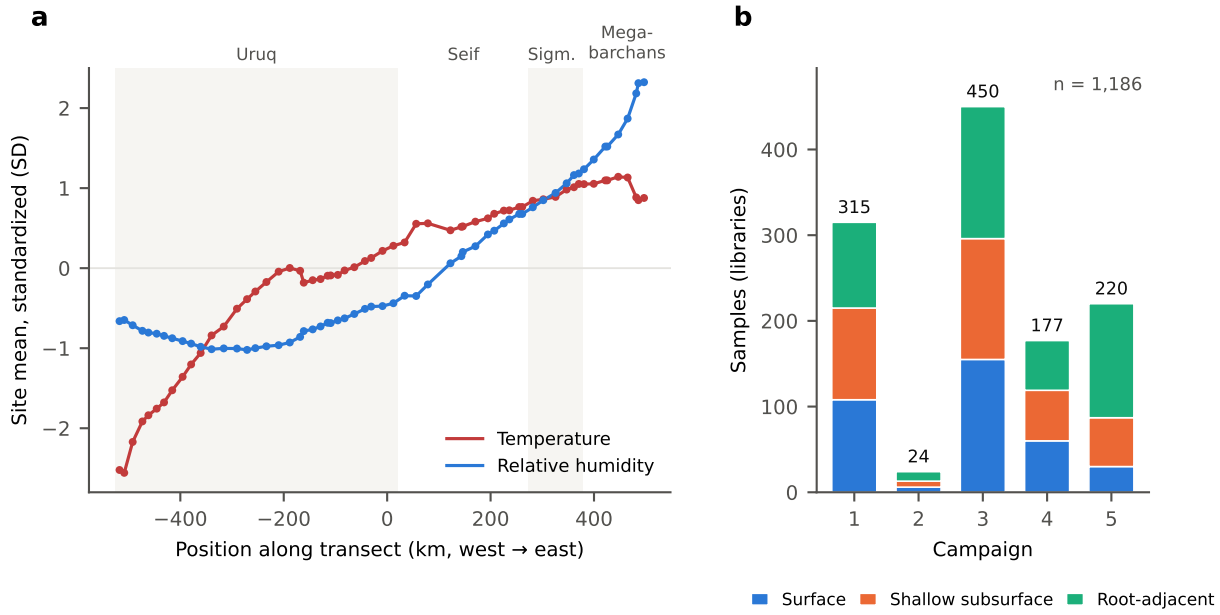


Figure S4: (a) Temperature and relative humidity from the 49-month period across the transect. (b) Number of samples used in the study across campaigns and compartments.

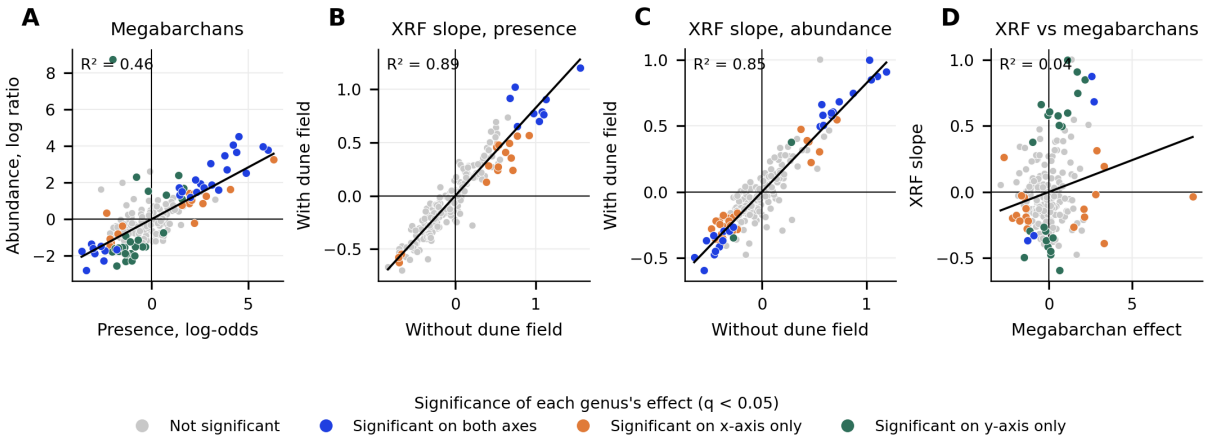


Figure S5: Comparison of genus effect sizes. Each point is one of the 200 genera, colored by whether its effect is significant ($q < 0.05$) on both axes, one axis or neither; lines are least-squares fits. (**A**) Megabarchans vs uruq: presence vs abundance where present. (**B**, **C**) Site-mean XRF slope without vs with dune field, for presence (B) and abundance (C). (**D**) Megabarchan effect vs XRF slope in the same abundance model. All effects are relative to the average genus. Both axes come from the same data, so R^2 is descriptive.

Table S7: Alpha diversity in each dune field relative to the uruq field (SPDE models without environmental covariates). Richness differences are percentages; Shannon (nats) and normalized Shannon differences are absolute. Posterior medians with 95 % credible intervals.

Dune field	Rarefied richness (%)	Shannon	Normalized Shannon
Seif	−7 (−23, 27)	−0.15 (−0.43, 0.21)	−0.008 (−0.030, 0.014)
Sigmoidal and hooked	−44 (−56, −15)	−0.52 (−0.87, 0.01)	−0.018 (−0.047, 0.011)
Megabarchans	−79 (−84, −68)	−1.65 (−2.03, −1.13)	−0.088 (−0.116, −0.059)

Table S8: Spatial structure of rarefied richness and the effect of adding dune field. Standard deviations are on the log scale. Prediction gain is the improvement in leave-site-out log score when dune field is added.

	Without dune field	With dune field
Spatial field SD	0.56	0.09
Site SD (not captured by the spatial field)	0.08	–
Spatial range (km, posterior median)	970	–
Prediction gain with dune field		
Rarefied richness	29.8	
Shannon	21.6	
Normalized Shannon	11.4	

Table S9: Between-site associations with alpha diversity, per standard deviation of the site-level covariate (SPDE models). Richness in percent; Shannon in nats. Climate associations are descriptive (Section S4.1).

Covariate and model	Rarefied richness (%)	Shannon
Elemental score	−13.5 (−21.0, −5.2)	−0.18 (−0.31, −0.04)
with dune field	−8.8 (−15.7, −1.2)	−0.09 (−0.21, 0.02)
without low diversity profiles	−5.4 (−12.6, 2.5)	−0.10 (−0.23, 0.03)
pH	−15.1 (−30.3, 0.3)	−0.24 (−0.51, 0.03)
Humidity	−38.6 (−50.3, −18.5)	−0.51 (−0.71, −0.22)
Temperature	−16.7 (−42.6, 23.5)	−0.32 (−0.69, 0.09)

Table S10: Differences in alpha diversity between compartments within sites, estimated from spatial (SPDE) models. Richness differences are percentages; Shannon (nats) and normalized Shannon differences are absolute. Posterior medians with 95 % credible intervals.

Contrast	Rarefied richness (%)	Shannon	Normalized Shannon
Root-adjacent vs surface	−9.9 (−17.2, −1.8)	−0.15 (−0.27, −0.03)	−0.029 (−0.039, −0.018)
Root-adjacent vs shallow subsurface	−17.7 (−24.3, −10.3)	−0.26 (−0.38, −0.15)	−0.041 (−0.052, −0.031)
Shallow subsurface vs surface	9.5 (0.5, 19.2)	0.11 (−0.01, 0.23)	0.012 (0.002, 0.023)

Table S11: Distance decay per compartment. Slopes are changes in dissimilarity per 100 km from pairwise regressions on the 1,770 site pairs. 95 % intervals from a 12-block jackknife along the transect.

	Surface	Shallow subsurface	Root-adjacent
Aitchison slope	1.35 (0.48, 2.21)	1.52 (0.57, 2.47)	1.18 (0.20, 2.15)
Mean Sørensen	0.41	0.42	0.41
Turnover share (%)	74	71	76
Sørensen slope	0.018 (−0.002, 0.039)	0.026 (−0.003, 0.054)	0.017 (0.002, 0.032)
Turnover slope	0.008 (−0.002, 0.018)	0.010 (−0.006, 0.026)	0.012 (0.000, 0.023)
Nestedness slope	0.011 (−0.004, 0.025)	0.016 (−0.017, 0.049)	0.005 (−0.005, 0.015)

Table S12: Aitchison distance decay with dune field. Upper rows: one model with distance and, for each dune field, the number of sites in the pair belonging to that field (0–2; uruq as reference); field coefficients are the added dissimilarity per such site. Bottom row: a separate model with distance and an indicator for pairs from different fields. 95 % intervals from a 12-block jackknife.

	Surface	Shallow subsurface	Root-adjacent
Distance (per 100 km)	1.19 (0.74, 1.63)	1.23 (0.67, 1.79)	0.88 (0.17, 1.60)
Seif	1.9 (−3.3, 7.0)	1.2 (−0.2, 2.6)	0.8 (−0.9, 2.4)
Sigmoidal and hooked	2.4 (0.9, 4.0)	1.5 (−4.8, 7.8)	3.0 (−3.9, 9.8)
Megabarchans	5.0 (−2.0, 12.1)	7.3 (0.5, 14.1)	6.9 (1.0, 12.9)
Different dune fields	3.9 (−1.8, 9.6)	4.0 (−0.6, 8.7)	4.3 (−0.1, 8.6)

S7 Genome-resolved integration

S7.1 Cohorts, functional calls and contrasts

The refined genome catalogue (EQ-MAG-REFINED-v1.0.0) contains 5,438 species representatives: 126 high-quality and 5,311 strict medium-quality genomes (< 10% contamination) under MIMAG standards, plus one boundary representative (48Dr2__self_SemiBin_25) at 64.61% completeness and 10.0% contamination. From 551 read-bearing shotgun libraries (EQ-SHOTGUN-READ-LIBS-v1.0.0, 554 paired-end sequencing runs across 221 Illumina NovaSeq 6000 and 333 Illumina NovaSeq X runs), abundance analyses (EQ-COVERM-REL-ABUND-v1.0.0) retained 546 core-transect profiles across 59 sites (489 unique specimen keys; EQ-ECOL-SHOTGUN-CORE-546-v1.0.0) after excluding one library with zero mapped read share (F5Dr1, 100% unmapped; leaving 550 mapped libraries) and four out-of-core libraries from Sites 62 and 64 (62PRr2, 64Dr3, 64PRr2 and 64Sr2). Exact specimen matching to the 1,186 retained amplicon libraries yielded 442 profiles across 58 sites (EQ-ECOL-MATCHED-442-v1.0.0). A missing read-depth value in the spatial covariate table (F45PRr3) reduced the spatial-model cohorts to 545 (EQ-ECOL-SHOTGUN-MODEL-545-v1.0.0) and 441 (EQ-ECOL-MATCHED-MODEL-441-v1.0.0), respectively. These cohorts differ from the marker-read screening set described below because genome abundance and marker-read analyses have separate quality filters.

We searched 20,451,986 representative-genome proteins called by Pyrodigal (EQ-PROT-PYRODIGAL-REP-v1.0.0) with 106 published microTrait HMMs and evaluated 41 rules at trusted cutoffs [23]. A separate KO projection used the frozen analytical representative-genome KO matrix (rep_ko_v2.tsv, EQ-KO-LEGACY-ANALYTICAL-v0; IBEX 04_functional/repko_clean/results) to test annotation dependence; neither EQ-PROT-PYRODIGAL-REP-v1.0.0 nor rep_ko_v2.tsv is asserted to derive from the foundational 19,186,792-protein Bakta v1.12.0 and eggNOG-mapper v2.1.13 release annotation product (EQ-PROT-BAKTA-v1.0.0; [PENDING: DATA-PROVENANCE-UNVERIFIED-PROTEIN-EQUIVALENCE]). A genome's rule call was weighted by its abundance either within the mapped community or relative to the original read pool; these denominators were not interchanged. The higher-quality sensitivity retained genomes with at least 70% completeness and at most 5% contamination. Surface–subsurface contrasts used 68 visits at 36 sites. Replicates were averaged within visit and compartment; paired differences were then averaged within site, with intervals from resampling ten contiguous geographic blocks.

For each genus shared between compartments, the change in trait abundance was decomposed exactly into the change in genus abundance multiplied by its mean carrier fraction and the change in carrier fraction multiplied by its mean abundance. Groups detected in only one compartment supplied a separate exclusive component. By algebra, the total compartment contrast is invariant to taxonomic assignment, whereas genus labels partition that total across the between-shared, within-shared and exclusive components. Genomes without a named genus assignment (g__) entered as individual singleton groups rather than a pooled bin; they are not treated as independent biological lineages, and their within-group component is zero by construction because within-lineage change cannot be estimated for singletons. Under the frozen analytical taxonomy snapshot (EQ-TAX-LEGACY-ANALYTICAL-v0, genome_quality.tsv) retained in the frozen 41-trait matrix (Table S13), unclassified-genus genomes accounted for a mean 27.5% of mapped abundance in the paired surface–subsurface comparison (26.04% under canonical gtdb_r232_classification, and 23.86% versus 25.72% in the paired root-adjacent–subsurface comparison). Whereas total community trait contrasts and bacterial MAG Hill diversity do not depend on below-domain taxonomy labels (domain assignments agree across all 5,438 representatives), 1,898 full taxonomy strings (including 247 phylum, 34 class, 220 order and 1,124 genus labels) differ between EQ-TAX-LEGACY-

ANALYTICAL-v0 and canonical gtdb_r232_classification in EQ-MAG-REFINED-v1.0.0. Recomputing the exact decomposition under canonical gtdb_r232_classification for the three focal guilds (BetL, AhpCF and group 1l [NiFe]-hydrogenase; Table S14) confirmed that between-genus and compartment-exclusive replacement dominate each contrast while within-genus changes remain small; non-focal traits in Table S13 retain their frozen legacy decomposition components without a canonical replay. The three-component decomposition reported here differs from an earlier trace-gas summary that combined exclusive and between-genus contributions.

BetL and AhpCF surface enrichment retained the same direction under both annotation approaches, both denominators, the higher-quality restriction and matched-cohort analysis. The complete panel showed different responses among stress-protection and DNA-repair capacities (Table S13). Photolyase, carotenoid and mismatch-repair markers were more represented at the surface, whereas superoxide dismutase and several baseline repair pathways had uncertain contrasts. Betaine biosynthesis and other osmolyte-synthesis markers provided weaker or annotation-dependent evidence than uptake. Trehalose illustrates this dependence: the microTrait HMM rule gave a positive surface contrast, while its KO projection and the stricter two-step synthesis rule did not. The stage-II sporulation protease marker spoIIGA alone does not establish a complete sporulation pathway. Table S13 reports paired estimates, spatial-model intervals, multiplicity adjustments and turnover components. The reproducibility package supplies the complete panel-by-contrast matrices, including unavailable and annotation-sensitive entries.

S7.2 Spatial models and numerical audit

We fitted standardised responses with compartment (reference: shallow subsurface), campaign and standardised log read depth as fixed effects, optionally including dune field (reference: uruq). The hierarchy included site, site–visit and site–visit–compartment random intercepts and a two-dimensional Matérn SPDE field. The primary specification used a Student-t likelihood with ten degrees of freedom and a 200-km spatial-range prior; sensitivity fits used 50- and 500-km range priors, a Gaussian likelihood, the matched cohort and the higher-quality genome set. Diversity responses were log-transformed; marker responses used $\log_2(\mathrm{GE}+0.01)$. The range prior assigned probability 0.5 below its stated range; the spatial standard-deviation prior assigned probability 0.01 above one standardised response unit. Random-intercept and residual precision priors likewise used PC priors with parameters $(1, 0.01)$. Fixed effects had unit precision on the standardised scale and a flat intercept. The mesh used inner and outer maximum edges of 40 and 200 km, a 5-km cutoff and 50- and 300-km extensions. The primary coefficients below are transformed back from response standardisation. Chemistry covariates were unavailable in these integration inputs and were not fitted in the functional models.

The screen produced 1,800 fits across 207 responses; 29 requested response–specification combinations lacked a result. Six fits triggered numerical diagnostics (covariance conditioning, CPO failure or mode flags): one KO-projected ammonia-oxidation fit, three glucosylglycerol fits and two whole-read-denominator MAG group-1l fits. Those fits were excluded from evidential support. In particular, the group-1l conclusions use the independently quantified marker reads, not the flagged MAG fits. Complete fit coefficients, diagnostics and analysis scripts are provided in Supplementary Data 1. Intervals in this broad Bayesian screen were not multiplicity-adjusted; headline results were checked against paired analyses and sensitivity fits. Missing or numerically flagged results do not provide evidence for an absent association.

As an exploratory sensitivity analysis complementing the separate primary INLA posterior fits in Table S15, we evaluated the paired within-sample difference in the eastern megabarchan versus uruq slope ($\beta_{\mathrm{tax}} - \beta_{\mathrm{KO}}$) using covariate-adjusted linear fixed-effect regressions resampled jointly across

Table S13: Selected surface-minus-subsurface functional panel from the Stage 4 synthesis (`task_d_h6_baseline_vs_protection_surface_deep.tsv`). Paired site-block bootstrap estimates ($n = 68$ visits across 36 sites; 10 spatial blocks) and primary INLA spatial-model posterior estimates ($n = 545$ libraries across 59 sites) are reported in percentage points of the mapped community, alongside the frozen three-component genus turnover decomposition under the legacy analytical taxonomy snapshot (`EQ-TAX-LEGACY-ANALYTICAL-v0`; canonical `gtdb_r232_classification` decompositions for BetL, AhpCF and group 1l are reported in Table S14, whereas non-focal traits were not replayed). KO rows use the frozen analytical representative-genome KO matrix (`rep_ko_v2.tsv`, `EQ-KO-LEGACY-ANALYTICAL-v0`). Paired q-values are Benjamini–Hochberg adjustments of site-level Wilcoxon signed-rank tests (q_{all41} across the 41-rule family; q_{fam} within each exploratory post-fit family). Their dependence assumptions differ from those of the spatial-block intervals; spatial-model estimates and intervals guide the habitat interpretation. [a]HMM and KO annotations give different conclusions; [b]a single sporulation marker rather than a complete pathway.

Panel	Family	Paired [95% CI]	q_{all41} / q_{fam}	INLA [95% CrI]	Betw.	With.	Excl.
`HMM:glycine betaine transporter` (BetL)	Osmolyte/stress	$+7.55$ $[5.11, 10.16]$	2.1×10^{-5} / 1.5×10^{-5}	$+6.70$ $[5.24, 8.16]$	$+6.12$	-0.21	$+1.65$
`KO:glycine betaine transporter`	Osmolyte/stress	$+12.17$ $[7.71, 16.25]$	1.9×10^{-5} / 1.2×10^{-5}	$+11.23$ $[8.15, 14.27]$	$+8.40$	$+0.39$	$+3.39$
`HMM:choline/glycine/proline betaine`	Osmolyte/stress	$+6.87$ $[4.33, 10.00]$	1.2×10^{-4} / 1.0×10^{-4}	$+5.45$ $[3.08, 7.80]$	$+3.22$	$+0.64$	$+3.01$
`HMM:mechano-sensitive channel`	Osmolyte/stress	$+6.88$ $[4.02, 10.27]$	1.7×10^{-5} / 1.4×10^{-5}	$+5.95$ $[4.09, 7.80]$	$+5.87$	$+0.26$	$+0.74$
`HMM:betaine_biosynthesis`	Osmolyte/stress	-1.22 $[-3.27, 0.96]$	0.341 / 0.345	-0.76 $[-2.61, 1.09]$	-0.65	-0.28	-0.28
`KO:betaine_biosynthesis`	Osmolyte/stress	$+3.25$ $[0.66, 6.72]$	0.086 / 0.099	$+3.38$ $[1.26, 5.48]$	$+3.08$	-0.63	$+0.80$
`curated:ectoine_marker`	Osmolyte/stress	$+0.76$ $[-1.27, 3.44]$	0.798 / 0.775	-0.39 $[-2.45, 1.65]$	$+0.59$	-0.15	$+0.32$
`HMM:trehalose_biosynthesis`[a]	Osmolyte/stress	$+12.07$ $[7.35, 17.26]$	1.2×10^{-4} / 1.0×10^{-4}	$+11.47$ $[8.23, 14.67]$	$+8.61$	$+0.15$	$+3.31$
`KO:trehalose_biosynthesis`[a]	Osmolyte/stress	-1.51 $[-4.94, 3.49]$	0.346 / 0.336	-0.66 $[-2.84, 1.52]$	-0.25	$+0.38$	-1.64
`curated:trehalose_strict`[a]	Osmolyte/stress	$+0.20$ $[-3.03, 4.97]$	0.863 / 0.830	$+0.78$ $[-1.21, 2.78]$	$+1.29$	$+0.26$	-1.34
`HMM:spoIIGA`[b]	Osmolyte/stress	$+0.63$ $[-0.10, 1.59]$	0.093 / 0.112	$+0.27$ $[-0.10, 0.64]$	$+0.84$	-0.21	$+0.01$
`HMM:ahpCF`	UV/oxidative	$+6.95$ $[4.04, 9.65]$	1.5×10^{-5} / 1.4×10^{-5}	$+5.53$ $[3.84, 7.22]$	$+5.28$	$+0.29$	$+1.38$
`KO:ahpCF`	UV/oxidative	$+5.79$ $[3.36, 8.01]$	5.2×10^{-5} / 2.7×10^{-5}	$+4.64$ $[3.16, 6.13]$	$+4.24$	$+0.11$	$+1.44$
`HMM:sodA`	UV/oxidative	-0.32 $[-3.63, 2.87]$	0.966 / 0.895	-0.47 $[-2.59, 1.66]$	$+0.24$	-0.22	-0.34
`KO:sodA`	UV/oxidative	-1.13 $[-4.04, 1.82]$	0.463 / 0.408	-1.17 $[-3.28, 0.93]$	-0.52	-0.26	-0.35
`legacy:photolyase`	UV/oxidative	$+10.14$ $[4.88, 15.04]$	0.0012 / 7.9×10^{-4}	$+11.36$ $[8.95, 13.75]$	$+6.31$	$+1.02$	$+2.81$
`legacy:carotenoid`	UV/oxidative	$+9.95$ $[6.23, 14.02]$	7.9×10^{-5} / 3.3×10^{-5}	$+10.49$ $[8.23, 12.76]$	$+5.31$	$+1.04$	$+3.61$
`HMM:lycopene biosynthesis`	UV/oxidative	$+4.98$ $[1.47, 8.55]$	0.034 / 0.029	$+6.34$ $[4.11, 8.57]$	$+2.28$	$+0.77$	$+1.93$
`legacy:mismatch_repair`	Baseline repair	$+8.91$ $[3.97, 12.27]$	0.0017 / 0.0014	$+8.17$ $[5.15, 11.16]$	$+5.74$	$+0.10$	$+3.07$
`legacy:base_excision_repair`	Baseline repair	-0.07 $[-0.20, 0.07]$	0.245 / 0.164	-0.01 $[-0.08, 0.06]$	$+2.12$	-0.02	-2.16
`legacy:nucleotide_excision_repair`	Baseline repair	$+0.20$ $[-0.05, 0.51]$	5.2×10^{-5} / 2.7×10^{-5}	$+0.11$ $[0.06, 0.15]$	$+2.27$	-0.01	-2.06
`legacy:SOS_translesion`	Baseline repair	$+3.86$ $[0.40, 8.48]$	0.015 / 0.0097	$+1.08$ $[0.44, 1.71]$	$+4.86$	$+0.04$	-1.04
`legacy:nhej`	Baseline repair	-4.49 $[-7.69, 0.67]$	0.017 / 0.0097	-4.44 $[-6.52, -2.36]$	-1.47	-0.25	-2.76

Table S14: Genus-turnover decomposition for the three focal guilds under GTDB release 232 versus the earlier analytical taxonomy snapshot, in percentage points of the mapped community. Surface–subsurface contrasts span 36 sites and 68 paired visits; the root-adjacent–subsurface contrast spans 47 sites and 106 paired visits. Confidence intervals (95% CI) resample ten contiguous spatial blocks (1,000 draws). By algebra, Between + Within + Exclusive = Total, so total contrasts are identical across taxonomies. Genomes without a named genus (26.04% versus 27.52% of mapped abundance in surface–subsurface pairs, and 23.86% versus 25.72% in root-adjacent–subsurface pairs) enter as singletons, are not independent lineages, and have zero within-group change by construction.

Guild/contrast	Component	Canonical [95% CI]	Earlier estimate
BetL (Surface − Subsurface)	Total	+7.55 [4.72, 10.33]	+7.55
	Between shared	+5.89 [4.01, 7.68]	+6.12
	Within shared	−0.15 [−0.83, 0.39]	−0.21
	Exclusive	+1.80 [0.99, 2.78]	+1.65
AhpCF (Surface − Subsurface)	Total	+6.95 [3.34, 9.71]	+6.95
	Between shared	+5.28 [2.05, 8.00]	+5.28
	Within shared	+0.22 [−0.03, 0.48]	+0.29
	Exclusive	+1.45 [0.71, 2.25]	+1.38
Group 1l (Root-adjacent − Subsurface)	Total	−18.77 [−25.25, −11.82]	−18.77
	Between shared	−11.57 [−15.46, −7.77]	−11.35
	Within shared	−0.41 [−0.59, −0.21]	−0.58
	Exclusive	−6.79 [−9.65, −3.60]	−6.84

Table S15: Selected coefficients from the primary spatial model ($n = 545$ libraries across 59 sites), with 95% credible intervals. Surface and root-adjacent contrasts use shallow subsurface as reference; eastern contrasts compare megabarchan with uruq fields. Percentage-point effects refer to the mapped community; annotated KO diversity uses the frozen analytical KO matrix (`rep_ko_v2.tsv`, `EQ-KO-LEGACY-ANALYTICAL-v0`).

Response	Contrast	Estimate [interval]
Bacterial MAG Hill diversity (log)	Eastern field	−1.132 [−1.589, −0.327]
Annotated KO Hill diversity (log)	Eastern field	0.025 [−0.022, 0.073]
BetL (percentage points)	Surface	6.70 [5.24, 8.16]
AhpCF (percentage points)	Surface	5.53 [3.84, 7.22]
Group 1l excess ($\log_2$ units)	Root-adjacent	−1.733 [−2.038, −1.424]
Group 1h excess ($\log_2$ units)	Root-adjacent	−0.083 [−0.213, 0.047]
Group 2a excess ($\log_2$ units)	Root-adjacent	0.052 [−0.083, 0.187]
Form-I CoxL excess ($\log_2$ units)	Root-adjacent	−0.117 [−0.224, −0.011]

the ten spatial blocks (`task_d_eastern_joint_diversity_contrast.tsv`; Table S16). Because the eastern megabarchan sites are concentrated in the eastern spatial blocks, 315 of 1,000 sample-level block resamples (233 of 1,000 for site–compartment averages) omitted the target dune group and were rank-deficient; the resulting intervals are therefore conditional on the 685 (or 767) identifiable resamples and are reported as conditional sensitivity summaries rather than primary INLA posterior contrasts or formal equivalence tests. Across the matched 441-sample cohort (58 sites; 27 megabarchan and 308 uruq units) and the 545-sample shotgun cohort (59 sites; 33 megabarchan and 363 uruq units), bacterial MAG versus KO log-$q = 1$ slope differences were -1.280 (conditional 95% block CI -1.503 to -1.064) and -1.309 (-1.507 to -1.050) under site-balanced weighting, -1.550 (-1.767 to -1.214) and -1.506 (-1.669 to -1.211) under sample weighting, and -1.372 (-1.489 to -1.222; $n = 149$) and -1.403 (-1.519 to -1.178; $n = 155$) for site–compartment averages. Standardized z-score slope differences ranged from -1.186 to -1.425 across specifications. Amplicon 16S versus KO copy-number diversity comparisons use the $n = 441$ ($n = 149$ site–compartment) matched subset across 58 sites where 16S Hill diversity is defined, yielding $\beta_{16S} - \beta_{KO} = -0.713$ (-1.087 to -0.491) under site-balanced weighting.

Table S16: Conditional joint spatial-block bootstrap sensitivity for the eastern megabarchan versus uruq diversity slope difference ($\beta_{tax} - \beta_{KO}$; `task_d_eastern_joint_diversity_contrast.tsv`). Estimates are covariate-adjusted linear fixed-effect slope differences in natural-log Hill $q = 1$ units and standardized z-score units, with 95% intervals conditional on identifiable spatial-block bootstrap draws (685/1,000 valid for sample-level fits; 767/1,000 for site–compartment averages). KO copy-number diversity uses the frozen analytical KO matrix (`rep_ko_v2.tsv`, `EQ-KO-LEGACY-ANALYTICAL-v0`). Amplicon 16S rows evaluate the $n = 441$ (149 site–compartment) matched subset across 58 sites.

Cohort	Pair	Weighting (n, sites)	β_{tax}	β_{KO}	$\Delta\beta_{log}$ [cond. 95% CI]	$\Delta\beta_z$ [cond. 95% CI]
Matched 441	MAG vs. KO	Site-balanced (441, 58)	−1.337	−0.056	−1.280 [−1.503, −1.064]	−1.186 [−1.674, −0.325]
Matched 441	MAG vs. KO	Sample-weighted (441, 58)	−1.564	−0.015	−1.550 [−1.767, −1.214]	−1.425 [−1.810, −0.805]
Matched 441	MAG vs. KO	Site–compartment (149, 58)	−1.416	−0.044	−1.372 [−1.489, −1.222]	−1.391 [−1.818, −0.630]
All 545	MAG vs. KO	Site-balanced (545, 59)	−1.333	−0.025	−1.309 [−1.507, −1.050]	−1.298 [−1.709, −0.594]
All 545	MAG vs. KO	Sample-weighted (545, 59)	−1.503	+0.002	−1.506 [−1.669, −1.211]	−1.421 [−1.741, −0.927]
All 545	MAG vs. KO	Site–compartment (155, 59)	−1.425	−0.022	−1.403 [−1.519, −1.178]	−1.387 [−1.747, −0.785]
Matched 441	16S vs. KO	Site-balanced (441, 58)	−0.769	−0.056	−0.713 [−1.087, −0.491]	−0.533 [−1.181, +0.251]
Matched 441	16S vs. KO	Sample-weighted (441, 58)	−0.996	−0.015	−0.981 [−1.372, −0.578]	−0.878 [−1.332, −0.323]
Matched 441	16S vs. KO	Site–compartment (149, 58)	−0.793	−0.044	−0.749 [−1.097, −0.506]	−0.678 [−1.201, +0.068]

S7.3 PICRUSt2 comparison and archaeal mapping controls

PICRUSt2 and MAG-derived KO profiles (`rep_ko_v2.tsv`, `EQ-KO-LEGACY-ANALYTICAL-v0`) were normalised to each platform's complete KO total, not reclosed to the shared KO subset. On the 442 exact matched profiles across 58 sites, 8,860 KOs were shared. Replicates were averaged within visits and paired visit differences within sites. Spearman correlations of the compartment effects were 0.520 (surface; 95% block interval 0.305–0.545) and 0.552 (root-adjacent; 0.432–0.569), with direction agreement of 67.0% and 67.8%. Intervals used 1,000 joint resamples of ten contiguous blocks. Restricting to 6,678 prevalent KOs increased correlations to 0.576 and 0.590 and direction

agreement to 71.2% and 72.4%. Agreement is partial and does not validate individual PICRUSt2 predictions.

Competitive archaeal mapping (`EQ-COVERM-ARCHAEA-COMP-v1.0.0`) included all 5,438 representatives and 301 external Thermoplasmatota genomes, requiring at least 95% read identity and 75% aligned read length. The paired review covered 193 profiles, 68 visits and 36 sites. At a 20% genome-breadth threshold, 35 sites had a greater Thermoplasmatota read share in subsurface soil, with a mean excess of 0.528 percentage points (95% block interval 0.258–0.903). A 50% breadth threshold retained a positive contrast (0.472 points; 0.210–0.816). A MAPQ-20 analysis of unnormalised coverage (`EQ-COVERM-ARCHAEA-MAPQ20-v1.0.0`) had a wider interval spanning zero. These controls support the relative-abundance contrast under breadth filtering but do not establish an absolute abundance difference or activity.

S8 Trace-gas analysis: detailed methods and validation

Sampling, sequencing and genome catalogue

Sampling, DNA extraction, sequencing and the genome catalogue are shared with companion papers on the same transect (main Methods and the accompanying data resource [45]). Sixty sites along a 1,043 km west to east transect through the Saudi Rub' al-Khali were sampled on five expeditions (March 2023, July 2023, February 2024, August 2024 and October 2025). At each site and visit, three co-located replicates were taken at three positions: surface soil (0 cm), bulk subsurface soil (5–10 cm) and soil from the root zone of the nearest plant at 5–10 cm, which we call root-adjacent because soil was not consistently attached to roots. Soil was frozen at −20°C in the field and stored at −80°C. DNA was extracted from 0.9–1.2 g of soil with the QIAGEN PowerLyzer PowerSoil or DNeasy PowerSoil Pro kit and sequenced on Illumina NovaSeq 6000 (221 paired-end runs) and Illumina NovaSeq X (333 paired-end runs) instruments across 554 runs from 551 read-bearing libraries (150-nt paired reads; 101-nt reads for the fourth expedition). Reads were quality-filtered with fastp v1.0.1 [9]. Samples from two special-interest locations outside the repeated transect (sites 62 and 64) and samples without site metadata were excluded, so the analysis covers sites 1–60.

Each sample was assembled with MEGAHIT v1.2.9 [27] and binned with SemiBin2 v2.2.1 [37]; bins were refined with DAS_Tool v1.1.7 [42], assessed with CheckM2 v1.1.0 [10], and the 12,434 bins of at least medium quality (completeness $\geq$ 50%, contamination $\leq$ 10%) were dereplicated with dRep v3.6.2 [34] at 95% average nucleotide identity into 5,438 species representatives (`EQ-MAG-REFINED-v1.0.0`: 126 high-quality and 5,311 strict medium-quality genomes with $<$ 10% contamination under MIMAG standards, plus one boundary genome, `48Dr2__self_SemiBin_25`, at 64.61% completeness and 10.0% contamination). In the foundational data release, representatives were classified with GTDB-Tk v2.7.2 [8] against GTDB release 232 [39] (`gtdb_r232_classification`) and annotated with Bakta v1.12.0 [41] and eggNOG-mapper v2.1.13 [7] (`EQ-PROT-BAKTA-v1.0.0`, 19,186,792 proteins); the three-guild genus-turnover decomposition was verified directly under canonical `gtdb_r232_classification` (Table S14), whereas the exploratory order- and genus-adjusted `gRodon2` growth analyses below used the frozen analytical taxonomy snapshot (`EQ-TAX-LEGACY-ANALYTICAL-v0`; [PENDING: `TAXONOMY-SNAPSHOT-SENSITIVITY`; confirm order- and genus-adjusted `gRodon2` regression using canonical assignments]). Relative abundance of each representative per sample was computed with CoverM v0.7.0 [2] in genome mode (`EQ-COVERM-REL-ABUND-v1.0.0`). For 1,010 representatives that refinement emitted as contig subsets of a parent bin, the Bakta/eggNOG release annotations were taken from the parent bin and restricted to genes on contigs retained in the representative, matched by sequence; genes on removed contigs were not counted towards carriage or traits.

Typing of hydrogenase and carbon monoxide dehydrogenase large subunits

Proteins annotated with the [NiFe]-hydrogenase large subunit orthologue (KEGG K06281) and the carbon monoxide dehydrogenase large subunit orthologue (K03520) were extracted from the representatives. Each hydrogenase was assigned the class of its best DIAMOND blastp hit in ultra-sensitive mode in HydDB [43, 18]. We analyse groups 1h, 1l and 2a separately, because all three include enzymes that oxidise atmospheric hydrogen [18, 22]. Each CoxL protein was called form I when it carried the form-I active-site motif AYXCSFR, including the variants with S in the first or L in the sixth position reported among form-I references [21], and form II when it carried AYRGAGR [25, 30, 12]; proteins without either motif were called form I at more than 55% identity and at least 90% query coverage to the three form-I references of Cordero et al. [11] and

were otherwise unresolved. All 59 form-I calls made without the motif were fragments in which the alignment did not span the motif region. A representative carries a group when it encodes at least one retained protein of that group.

To check the genomic context of typed genes, we took the ten genes on each side of every typed large subunit on the same contig from the Bakta annotation and scored the co-occurrence of the hydrogenase small subunit (K06282, K05927, K00437), hydrogenase maturation genes (K04651–K04656, K03605, K03192, or Bakta product names) and, for CoxL, the medium and small subunits (K03519, K03518) and the accessory genes *coxDEFG*. Proportions are reported for loci with at least five genes on both sides; they describe neighbourhood co-occurrence within the stated window.

For display, hydrogenase large subunits of at least 400 residues were clustered at 70% identity with MMseqs2 v18.8cc5c [44], combined with all HydDB group 1 and group 2 references, aligned with MAFFT v7.526 [24] (`--auto`), stripped of columns with more than 50% gaps and used to build a tree with FastTree v2.2.0 [40] (`-lg -gamma`), rooted on the clade that best separates the group 2 references. CoxL proteins of at least 600 residues were treated in the same way with the form-I references.

Read-based quantification

Groups were quantified in reads so that sequences outside the genome catalogue contribute. A competitive protein database contained every typed hydrogenase large subunit of the representatives labelled with its HydDB class, all HydDB reference hydrogenases, every CoxL protein of the representatives labelled form I, form II or unresolved, the three form-I CoxL references, and 14 universal single-copy ribosomal proteins of the representatives (KEGG K02886, K02906, K02926, K02931, K02933, K02867, K02874, K02878, K02982, K02988, K02994, K02946, K02948 and K02965) clustered at 90% identity. Every read is assigned to its single best hit, so related enzymes of other classes compete for each read. From each sample, 10 million forward reads were drawn with seqtk v1.5 [28] (seed 11) and truncated to 100 nt so that the 101-nt and 150-nt libraries were screened identically; reads shorter than 75 nt were discarded. Reads were searched with DIAMOND v2.0.15 [6] blastx (`--id 50 --evalue 1e-5 --max-target-seqs 1`), and each read's best hit with an alignment of at least 25 residues was counted at identity floors of 50, 60 and 70%. For each sample and label, reads were divided by the length of the hit subject in kilobases and summed, and each marker was divided by the median of the 14 ribosomal-protein values. The result, genome equivalents (GE), is an estimated ratio of marker to single-copy-gene coverage, which approximates marker copies per genome [32]; differences in read assignment between markers and the ribosomal proteins offset it from true copies per genome (see the mock communities below). GE describe relative gene representation among the genomes present. The primary measure used the 60% identity floor and samples in which at least 10 of the 14 ribosomal proteins were detected.

We report six markers: group 1l, 1h and 2a hydrogenases, form-I CoxL, all other [NiFe]-hydrogenases and all other CoxL (form II and unresolved). The last two serve as controls. Of the 573 sequenced samples screened, 546 from sites 1–60 had complete metadata and passed the ribosomal-protein filter (116 surface, 164 bulk and 266 root-adjacent samples; median 9.77 million reads screened).

To estimate how well reads are assigned when their source is absent from the database, we tiled 100-nt fragments every 30 nt over each typed gene and held out, in five folds each, (i) all marker proteins of 20% of the representatives, (ii) 20% of the clusters of catalogue marker proteins at 50% identity, with the HydDB and CoxL references retained, and (iii) 20% of graph components, obtained by single-linkage grouping of all marker proteins of the database, catalogue and references alike, at 70% identity over at least half of the shorter protein in a DIAMOND search ($E \leq 10^{-5}$),

with every member of a held-out component removed. Fixed identity thresholds can assign reads of one enzyme class to a homologue [35], and scheme (iii) tests removal of connected reference groups. A more permissive verification search ($E \leq 10$) still found 34 of 6,478 held-out proteins with matches above 70% identity at half-length coverage (maximum 83.3%); this benchmark therefore does not guarantee identity-disjoint references. At a 50% linkage threshold, groups 1l and 1h formed a single component, and holding out that component reduced recall to near zero. Fragments of 50 random non-marker genes per held-out genome were added as negatives, and fragments were classified exactly as the reads. To calibrate GE, we simulated 20 communities from the CoverM profiles of bulk and root-adjacent samples of ten site visits, drawing 4 million 150-nt reads with 0.5% substitutions from the representative genomes in proportion to their abundance and truncating them to 100 nt; true copies per genome were the cell-weighted mean number of typed markers. The reads were screened against the full database and against a database from which every marker and ribosomal protein of the community's own genomes had been removed. The form-I CoxL misassignment rates of scheme (ii) were used to correct form-I and other CoxL GE by inverting the 2×2 assignment matrix, as a sensitivity analysis.

Contamination screen

Representatives whose GTDB genus matched a 16S rRNA amplicon variant flagged in 17 extraction blanks of the fifth expedition or six of the fourth expedition were flagged, and a re-implementation of the frequency method of decontam [13] compared abundance models independent of and inversely proportional to extracted DNA concentration, both over all samples with a single recorded concentration and after centring within each combination of expedition and extraction kit. Reads whose best hit belonged to a representative flagged by either method were removed in a sensitivity analysis. The shotgun no-template control of the fourth expedition contained no reads assigned to groups 1l, 1h or 2a or to form-I CoxL; its only marker reads came from an *Escherichia* genome carrying group 1c and 1d hydrogenases. The fourth-expedition screen compared six blanks with 95 biological profiles and yielded seven candidate variants (one-sided Fisher $p < 0.1$, detection in at least two blanks and higher prevalence in blanks). The three named genera, *Alcaligenes*, *Brevibacterium* and *Nesterenkonia*, matched 14 representatives, four additional to the previous exclusion set; four variants lacked a named genus. Candidate sequences and calls are retained with the analysis. The combined exclusion set contained 387 representatives, including two catalogue orphans already removed from the primary analysis. These cross-assay genus matches define a conservative sensitivity screen and do not establish contamination of individual MAGs. Genus, genome-trait and candidate carbon-fixation models were refitted after these exclusions; their sensitivity estimates use model standard errors without repeating the site bootstrap.

Environmental covariates

Daily precipitation for 1995–2025 was obtained from the ERA5 reanalysis [19] at 0.25° resolution through the Open-Meteo archive (`models=era5`). Long-term mean annual precipitation was computed for 1995–2024 and, as sensitivity, from the NASA POWER 2001–2020 climatology based on MERRA-2 [17] and from Open-Meteo's default blend of ERA5 with the ECMWF operational model. Mean annual precipitation from ERA5 ranged from 13 to 19 mm across sites; because reanalysis precipitation in Saudi Arabia agrees only moderately with gauges [47], we treat it as a ranking of sites rather than an absolute water balance. The 60 sites fall into 41 ERA5 grid cells. Antecedent precipitation was the sum over the 90 and 365 days before each site visit from ERA5 and, as sensitivity, from NASA POWER daily data (MERRA-2 based) and the blended product

(365 days, first four expeditions). Surface skin temperature was the NASA POWER daily mean (TS) over the 30 days before the centre date of each expedition, and soil moisture the SMAP surface value over the preceding 30 days. Vegetation greenness was the median MODIS normalised difference vegetation index (NDVI; MOD13Q1 and MYD13Q1 v061, 250 m [14]) over the 60 days before sampling. Extract DNA concentration was taken from the sample ledger for samples with a single positive recorded value (267 of 546 samples). Soil geochemistry was summarised from X-ray fluorescence measurements of the same sites by centred log-ratio principal components and by Ca, Cl and S. Soil organic carbon and water content were not measured on these samples.

Statistical analysis

Analyses were run in R v4.4.3 with lme4 v2.0.6 [3], vegan v2.7.5 [33] and ape v5.8.1 [38]. A site visit is one site sampled on one expedition. For paired contrasts between positions, each marker was summarised as the median over replicates within a site visit and the $\log_2$ ratio of the medians computed for each visit with both positions, with a pseudocount of 0.01 GE (results without a pseudocount and with 0.001 and 0.1 are reported). Confidence intervals were obtained by resampling sites with replacement (2,000 resamples), and tests are Wilcoxon signed-rank tests on site means of the visit values, so that the site is the independent unit. The excess change of a high-affinity marker is its $\log_2$ ratio minus the mean of the two control $\log_2$ ratios. A shared change in the ribosomal-protein denominator would shift all markers equally, and the excess removes it up to the effects of the pseudocount and of taking replicate medians separately for each marker. Mixed models of $\log_2$ GE with marker, position, their interaction and expedition as fixed effects and site, site visit and sample as random intercepts tested the interaction by likelihood ratio. The contribution of position to each marker was summarised as the gain in marginal R^2 when position is added to a model with expedition.

To relate group 1l to biomass, $\log_2$ GE was regressed on $\log_2$ extract DNA concentration with position, expedition and extraction kit as fixed effects and site and site visit as random intercepts; the DNA term was split into within-visit and between-visit components, the model was repeated within visits that used a single kit and within bulk soil, each with extraction kit as a fixed effect and site and site visit as random intercepts. If DNA concentration measured biomass with classical error of reliability 0.8 or 0.6, the slope would be attenuated by that factor, and we report the correspondingly rescaled slopes as a conditional sensitivity. Under a model in which group 1l copies per gram scale with biomass with exponent α, GE scale with exponent $\alpha - 1$. The root effect on group 1l was estimated with and without the DNA term in the same samples.

The change in the share of the mapped community held by group 1l carriers between root-adjacent and bulk soil was decomposed into a genus-turnover term (change in genus abundance at the mean carrier fraction of each genus) and a within-genus term (change in the carrier fraction of each genus at its mean abundance). Genera absent from one position contributed zero abundance there, and the carrier fraction of a genus present in only one position was taken from that position; the turnover share was bootstrapped over sites. The same decomposition was applied across bulk-soil sites against mean annual precipitation, with genus shares averaged over all visits to a site and absent genera counted as zero. Within the 81 genera that contain both carriers and non-carriers, the logit of the carrier share of each genus at each site, clipped to 0.005–0.995, was modelled on standardised log precipitation in a Gaussian mixed model weighted by the square root of the genus share, with genus and site as random intercepts; clipping at 0.001 and 0.02 is reported. To separate the number of carriers from the number of copies per carrier, the root contrast was computed for the abundance share of carriers (F), their copy-weighted share (M) and copies per carrier (C).

For growth strategy, the minimum doubling time of each representative was predicted from

codon usage bias with gRodon2 v2.7.3 [48] in partial mode from coding sequences of at least 240 nt with ribosomal-protein genes as the highly expressed set; these are predictions of maximal growth potential under the codon-usage model. Doubling time, completeness-adjusted genome size and GC content were compared between carriers and non-carriers in mixed models with order as a random intercept. The community-weighted mean log doubling time of a sample was the CoverM-weighted mean over representatives with an estimate, and its shift between root-adjacent and bulk soil was computed per site visit. The mapped abundance with a doubling-time estimate was a median of 97–99% per position (lowest 1.5%), and the analysis was repeated in samples with at least 80% and 90% coverage and with unestimated representatives set to the fastest or the slowest observed doubling time. The excess change of group 1l was regressed on this shift with site and expedition as random intercepts, and the intercept, which estimates the excess at zero shift, was bootstrapped over sites. Across genera with a mean bulk-soil share of at least 0.05% of the mapped community, the median visit-level $\log_2$ ratio between root-adjacent and bulk soil was regressed on median log doubling time and the fractions of representatives carrying groups 1l, 1h and 2a and form-I CoxL, with phylum as a fixed effect and weights equal to the square root of the bulk-soil share; the model was repeated for genera with a nonzero ratio, for genera with at least two representatives, for genera present in at least 25% or 50% of paired visits, with carriage computed among representatives at least 90% complete, and with order instead of phylum as a fixed effect, and the group 1l coefficient was bootstrapped over sites by recomputing the genus ratios from resampled visits. As a candidate screen for co-carriage of carbon fixation, genomes encoding all three Calvin–Benson–Bassham marker orthologues (RuBisCO large and small subunits, K01601 and K01602, and phosphoribulokinase, K00855) were compared between group 1l and 1h carriers in a binomial mixed model with order as a random intercept, with and without genome completeness and among genomes at least 90% complete.

Spatial associations were Spearman correlations of site medians with mean annual precipitation for each position, marker and precipitation product, with each site entered once and confidence intervals from a bootstrap over k-means spatial blocks of the site coordinates ($k = 4, 6, 8, 10$), Moran's I of the residuals with inverse-distance weights, and partial correlations controlling for latitude, longitude and elevation, for the first two geochemical principal components, for Ca, Cl and S, and for the first principal coordinate of community composition. A mixed model over all samples regressed $\log_2$ GE on standardised log precipitation with position and expedition as fixed effects and site and site visit as random intercepts. For change over time, log antecedent precipitation was split into a within-site component and a site mean, and $\log_2$ visit medians were modelled on both with expedition as a fixed effect and site as a random intercept; the within-site slope was bootstrapped over sites and scaled to the within-site interquartile range of precipitation, and an interval was compared with an exploratory bound of ± 0.5 $\log_2$(GE+0.01) units. This bound corresponds to a factor of $\sqrt{2}$ on the pseudocount-shifted GE scale; its ecological interpretation is weakest for sparse markers. The exposure scale is the pooled interquartile range of within-site deviations of log(1+precipitation). The within-site estimates were repeated with ERA5 and NASA POWER exposures, with a bootstrap over ERA5 grid cells instead of sites, leaving out each expedition in turn and with pseudocounts of 0.001 and 0.1. These intervals are conditional on the available climate grids and do not propagate local rainfall measurement error; bootstrap fits with finite estimates were retained without a full convergence audit. Because the surface exposure models rest on 68 visits in which temperature was largely determined by expedition, they were repeated leaving out each expedition. Turnover of composition was partitioned by PERMANOVA [1] on Bray–Curtis dissimilarities between visit-level profiles (mean relative abundance over the replicates of a site, expedition and position) at sites visited at least twice, with whole visit profiles permuted among expeditions within each site and position; the dispersion test used the same permutation design.

The Benjamini–Hochberg procedure [4] was applied within each declared family: the ten quantities (six markers and four excess changes) of each position contrast; the twelve spatial correlations (three positions by four groups) for each precipitation product; the twelve within-site slopes for each antecedent-precipitation exposure; and the twelve surface exposure models (four groups by three exposures). Genus, DNA, NDVI, genome-trait and carbon-fixation co-occurrence models are exploratory; their reported p values are unadjusted. Sensitivity variants assess dependence on analysis choices within these same data.

Sensitivity variants recomputed the position contrasts and spatial correlations at identity floors of 50 and 70%, without reads to contamination-flagged representatives, without reads to the 122 group 1l proteins typed below 50% identity or 80% coverage to their HydDB hit, with motif-less form-I CoxL calls treated as unresolved, with CoxL GE corrected for misassignment, without the fifth and without the fourth expedition. The excess of group 1l was also computed against each control separately.

Supplementary Results

Genus-model sensitivity and carrier genome traits

Across 289 genera, the fraction of a genus carrying group 1l predicted its depletion beside roots (-0.81 $\log_2$ units, SE 0.18; site bootstrap -1.12 to -0.52) with predicted doubling time, carriage of the other groups and phylum in the model, whereas carriage of group 1h and form-I CoxL did not (Fig. S6b). The association held among genera with a nonzero ratio (-0.92), with at least two representatives (-1.08), present in at least 25% or 50% of paired visits (-1.12, -1.18), with carriage counted among genomes at least 90% complete (-0.98) and with order instead of phylum as a fixed effect (-1.03, SE 0.22). With order in the model, predicted doubling time no longer predicted genus depletion (-0.05, SE 0.15), and carriage of group 2a also did (-1.68, SE 0.67; 134 carriers), although group 2a showed no excess depletion at the community level. Finer phylogenetic control therefore shifts weight from predicted growth potential to lineage. Vegetation greenness was negatively associated with group 1l in bulk soil (-0.80 $\log_2$ units per standard deviation of NDVI, SE 0.19) and at the surface (-1.17, SE 0.16) with mean annual precipitation in the model, and the excess change beside roots was smaller at greener sites (site medians, $\rho = 0.37$, 47 sites).

Group 1l carriers also differed in genome traits. Within orders, they had higher GC content than non-carriers (+0.7 percentage points) and genomes of similar size (-1.9%, $p = 0.08$), whereas group 1h, group 2a and form-I CoxL carriers had larger genomes (+7%, +11% and +17%). Across all genomes, carriers of group 1l only had a smaller median estimated genome size than genomes without high-affinity groups (3.78 against 3.97 Mb), consistent with the across-lineage comparison of trace-gas oxidisers along a northern-China aridity gradient [46]; that study used broader functional groups and a different growth-prediction method. Our genome-size difference did not persist within orders. In a candidate screen for carbon fixation, all three Calvin–Benson–Bassham marker genes were encoded by 18% of group 1l-only and 8% of group 1h-only carriers; the difference was not significant within orders (log-odds 0.73, SE 0.39, $p = 0.06$) but was after adjustment for genome completeness (0.96, $p = 0.02$) and among genomes at least 90% complete (28% against 9%; 1.29, SE 0.54, $p = 0.02$). RuBisCO co-occurs with high-affinity hydrogenases in desert genomes worldwide [26]. These comparisons describe candidate gene co-occurrence; sequence-form and pathway validation would refine their functional interpretation. Excluding contamination-flagged representatives retained the negative genus association for group 1l (281 genera; phylum-adjusted coefficient -0.77, SE 0.18; order-adjusted -0.96, SE 0.21). The completeness-adjusted carbon-fixation association also remained positive (log-odds 1.04, SE 0.43). With the expanded exclusion set, the median

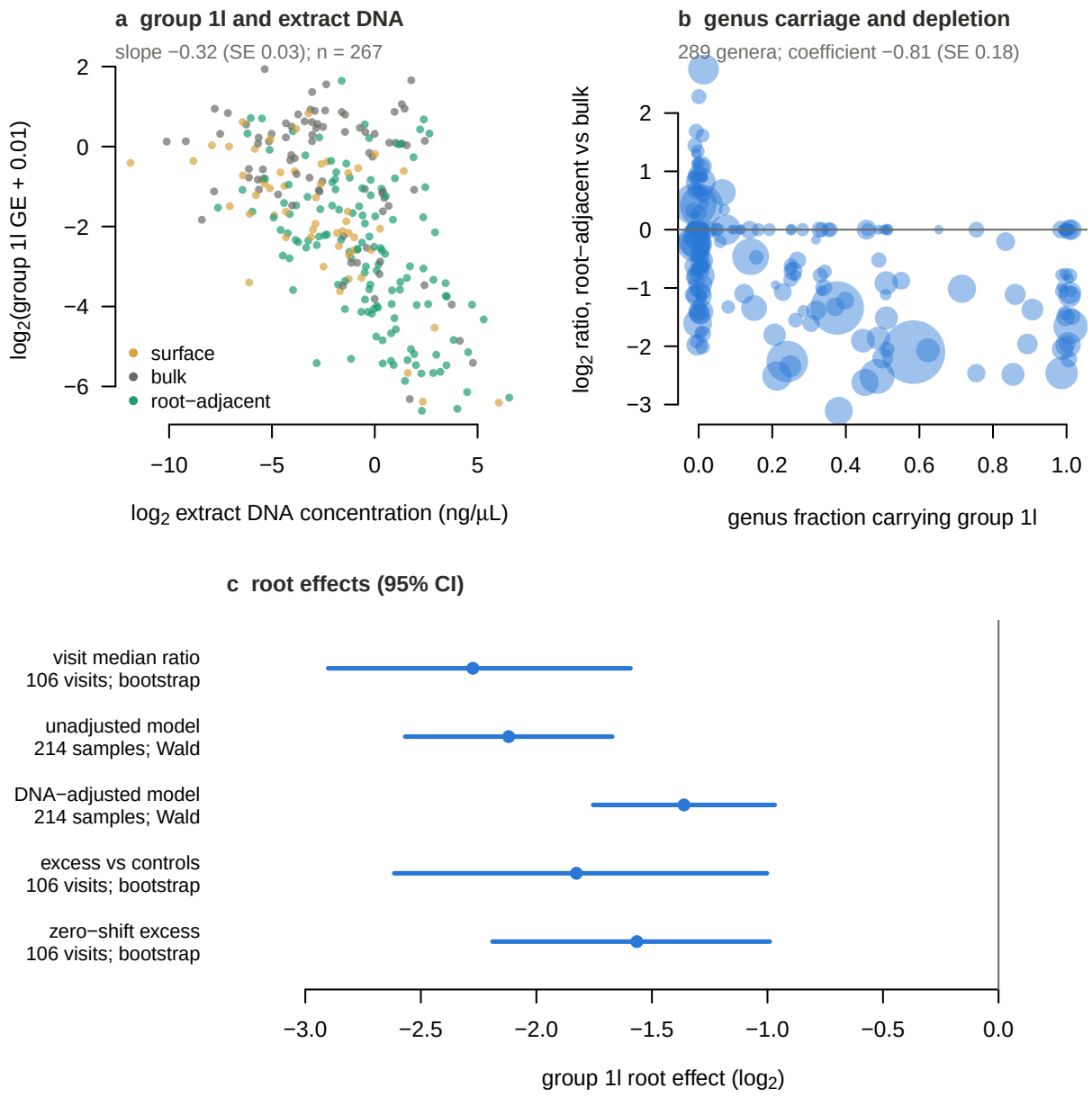


Figure S6: **The relative depletion of group 1l scales with DNA yield and with genus carriage, after adjustment for predicted growth potential. a**, Group 1l genome equivalents against extract DNA concentration, per sample, coloured by position, with the mixed-model slope annotated. **b**, Median $\log_2$ ratio of genus abundance between root-adjacent and bulk soil against the fraction of the genus's representatives carrying group 1l; symbol area scales with the genus's share of bulk soil; the coefficient is from a weighted regression with predicted doubling time, carriage of the other groups and phylum. **c**, Root effect on group 1l: the median visit $\log_2$ ratio, the mixed-model estimate without and with DNA concentration, the excess beyond the controls, and the excess predicted at zero shift in community growth potential, with 95% intervals.

root-to-bulk group-1l contrast was -2.29 $\log_2$ units and its excess was -1.76 (106 visits at 47 sites).

Spatial associations with aridity depend on the precipitation product

Across 51 sites, group 1l GE in bulk soil were negatively associated with mean annual precipitation from ERA5 ($\rho = -0.40$, $q = 0.007$), NASA POWER (-0.43) and the blended product (-0.51). The ERA5 association persisted after adjustment for geography (-0.36) and geochemistry (-0.30), but its spatial block bootstrap intervals included zero at every block count (six blocks: -0.64 to 0.07; Fig. S10). With POWER the intervals included zero at eight and ten blocks; with the blended product they excluded zero at all four block counts. Adjustment for community composition reduced the ERA5 correlation to -0.02.

At the surface, all four groups were negatively associated with ERA5 precipitation ($\rho = -0.46$ to -0.53, 45 sites), with spatial block intervals below zero. For group 1l, however, the correlation was weaker with POWER (-0.29, $q = 0.078$) and the blended product (-0.25, $q = 0.14$), whose block intervals included zero. Residual spatial autocorrelation remained in the ERA5 models for group 1l (Moran's $I = 0.16$, $p < 0.001$) and form-I CoxL ($p = 0.046$). These associations therefore depend on both the precipitation product and spatial structure.

The bulk-soil association also tracked lineage composition. Holding the carrier fraction of each genus at its mean reproduced the correlation of the carrier share with precipitation (-0.43 for both the observed and turnover-only shares). Within the 81 genera containing carriers and non-carriers, the Gaussian model of clipped logits gave a negative carrier-share slope (-0.29 per standard deviation of log precipitation, SE 0.12); changing the clipping threshold gave slopes of -0.38 and -0.21.

Within-site change between expeditions

The five expeditions span 31 months. In bulk soil, the within-site association of group 1l with precipitation over the preceding year was imprecise (103 visits at 35 sites; ERA5 $q = 0.34$). Resampling ERA5 grid cells gave an interval of -0.03 to 1.02 $\log_2$(GE+0.01) units across the within-site interquartile range of log precipitation, and leaving out individual expeditions gave estimates from -0.32 to 0.80 (Fig. S8).

At the surface and beside roots, group 1l was lower after wetter 90-day periods according to ERA5 (changes of -0.65 and -0.73 $\log_2$(GE+0.01) units; $q = 0.013$ and 0.020). These estimates remained negative under grid-cell resampling, alternative pseudocounts and omission of each expedition. With POWER exposures on the same visits, the estimates were -0.12 and $+0.33$, with intervals spanning zero. Within-site 90-day precipitation anomalies agreed weakly between products ($\rho = 0.19$ across 244 site visits), compared with 0.61 for annual anomalies. The seasonal association is therefore conditional on the precipitation product. Several other marker intervals lay within the exploratory ± 0.5 bound on $\log_2$(GE+0.01); their interpretation depends on the pseudocount and exposure range.

In the visit-level community profiles, expedition accounted for 2.1% of variation, site for 30.4% and position for 4.5%. Permuting visits within site and position gave $p = 0.001$ for expedition, with dispersion also differing ($p = 0.002$). Within bulk soil alone, expedition accounted for 3.7% ($p = 0.036$; dispersion $p = 0.26$). The community profiles document temporal change, while inference about rainfall responses is limited by the exposure products and the sparse repeated-site design.

Table S17: **Held-out benchmark of read assignment.** Fraction of 100-nt fragments from held-out genes assigned to their own group (recall) and fraction of fragments assigned to a group that came from it (precision), at the 60% identity floor, when all marker proteins of 20% of the representatives (genome) or 20% of catalogue 50%-identity clusters were removed with external references retained, or 20% of all-reference 70%-identity components were removed, over five folds. Component linkage required at least half-length coverage. A more permissive verification search found 34 held-out proteins with matches above 70% identity at this coverage, reaching 83.3%; the holdout is not strictly identity-disjoint.

Group	Recall	Precision	Recall	Precision	Recall	Precision
	Genome holdout		Catalogue cluster		All-reference component	
[NiFe] group 1l	0.98	1.00	0.89	1.00	0.66	0.92
[NiFe] group 1h	1.00	1.00	0.99	0.98	0.00	0.00
[NiFe] group 2a	1.00	1.00	0.94	1.00	0.23	0.98
Other [NiFe]	0.90	0.98	0.90	0.92	0.64	0.35
CoxL form I	0.97	0.92	0.82	0.77	0.44	0.85
Other CoxL	0.92	1.00	0.70	0.99	0.63	0.97

Table S18: **Calibration in 20 simulated catalogue communities.** True median marker copies per genome, median $\log_2$ GE bias, Pearson correlation on the log scale, and mean root-to-bulk $\log_2$ ratio error (number of evaluable pairs). Full: complete reference database. Removed: community genomes' marker and ribosomal proteins removed. Simulation assumes uniform coverage and catalogue-only genomes.

	True	Full			Removed		
Group	copies	Bias	r	Ratio error (n)	Bias	r	Ratio error (n)
[NiFe] group 1l	0.079	0.67	1.00	-0.11 (8)	0.79	1.00	-0.05 (8)
[NiFe] group 1h	0.038	0.80	0.97	0.27 (9)	0.95	0.94	0.40 (9)
[NiFe] group 2a	0.009	0.60	0.96	-0.16 (7)	0.46	0.94	-0.18 (7)
Other [NiFe]	0.000	6.25	0.65	0.93 (4)	6.39	0.63	0.95 (4)
CoxL form I	0.033	0.84	0.93	-0.10 (9)	1.22	0.90	-0.26 (9)
Other CoxL	0.697	1.11	0.98	0.03 (10)	1.04	0.97	0.16 (10)

Full microTrait annotation and focal surface contrasts

The complete microTrait pipeline processed all 5,438 canonical genomes and produced 37, 100 and 189 trait categories at increasing granularity. We used its exact BetL and AhpCF rule calls for a targeted sensitivity analysis of the two focal surface contrasts. BetL was positive in 341 genomes, all among the 415 positives in the earlier profile screen. AhpCF was positive in 522 genomes, all among the earlier 525 positives. The differences reflect the combined change in gene calling and profile processing; their separate contributions were not isolated. The surface contrasts retained positive spatial-block intervals (Table S21). The full profiles provide a broader reusable annotation product, while inference here remains focused on the stated ecological comparisons.

The complete release lacks the `aphA` and `phoN` profiles used as alternatives in the phosphatase rule. The coverage-aware output therefore retains 2,452 genomes as unassessed for absence of that trait; 2,986 have a positive assayed alternative. Neither BetL nor AhpCF depends on the missing profiles. Broad trait sums, carbon-fixation marker rules and predicted generation times were kept distinct from complete pathways, measured process rates and the specialised hydrogenase calls.

Table S19: **Sensitivity of the position contrasts and the spatial association.** Median $\log_2$ ratio of group 1l genome equivalents between root-adjacent and bulk soil, the excess beyond the controls with its site-level Wilcoxon p, the excess for group 1h and form-I CoxL, group 1l excess against other CoxL alone, the surface excess of group 1l, and the Spearman correlation of bulk-soil site medians of group 1l with ERA5 and NASA POWER mean annual precipitation.

Variant	Samples	1l ratio	1l excess	p	1l vs CoxL
Primary (60% identity)	546	-2.27	-1.83	4.9×10^{-7}	-1.81
50% identity	546	-2.29	-1.74	6.1×10^{-7}	-1.83
70% identity	546	-2.28	-1.85	2.9×10^{-7}	-1.79
Without expedition 5	455	-2.08	-1.54	1.5×10^{-6}	-1.60
Without expedition 4	461	-2.17	-1.69	1.2×10^{-5}	-1.71
Contamination-flagged removed	546	-2.29	-1.76	4.6×10^{-7}	-1.80
Weak group 1l calls removed	546	-2.25	-1.79	4.9×10^{-7}	-1.80
Motif-less form-I CoxL unresolved	546	-2.27	-1.82	4.9×10^{-7}	-1.80
CoxL misassignment-corrected	546	-2.27	-1.83	4.6×10^{-7}	-1.81

Variant	1h excess	Form-I excess	Surface 1l	ρ ERA5	ρ POWER
Primary (60% identity)	-0.01	-0.05	-0.83	-0.40	-0.43
50% identity	0.02	-0.01	-0.83	-0.40	-0.44
70% identity	-0.09	-0.16	-0.82	-0.40	-0.43
Without expedition 5	-0.09	-0.10	-0.83	-0.42	-0.48
Without expedition 4	0.06	-0.07	-0.88	-0.42	-0.41
Contamination-flagged removed	-0.03	-0.10	-0.82	-0.40	-0.43
Weak group 1l calls removed	-0.01	-0.05	-0.83	-0.40	-0.44
Motif-less form-I CoxL unresolved	-0.00	0.11	-0.83	-0.40	-0.43
CoxL misassignment-corrected	-0.01	-0.01	-0.83	-0.40	-0.43

Table S20: **Orders of group 1l carriers beside roots.** The ten orders of group 1l carriers with the highest mean share of the mapped community in bulk soil, with the median $\log_2$ ratio of their share between root-adjacent and bulk soil over 106 site visits and the fraction of visits with a lower share beside roots.

Phylum	Order	Bulk share (%)	Log_2 ratio	Visits lower
Actinomycetota	Acidimicrobiales	9.27	-2.83	83%
Actinomycetota	Solirubrobacterales	6.03	-1.72	79%
Actinomycetota	UBA4738	2.14	-2.65	71%
Actinomycetota	Rubrobacterales	1.17	-2.62	77%
Chloroflexota	Chloroflexales	1.15	-1.21	74%
Actinomycetota	Mycobacteriales	1.01	-0.67	61%
Gemmatimonadota	Longimicrobiales	0.94	-1.89	81%
Actinomycetota	CADDZG01	0.85	-2.49	72%
Actinomycetota	Gaiellales	0.82	-2.70	80%
Gemmatimonadota	Gemmatimonadales	0.77	-1.96	77%

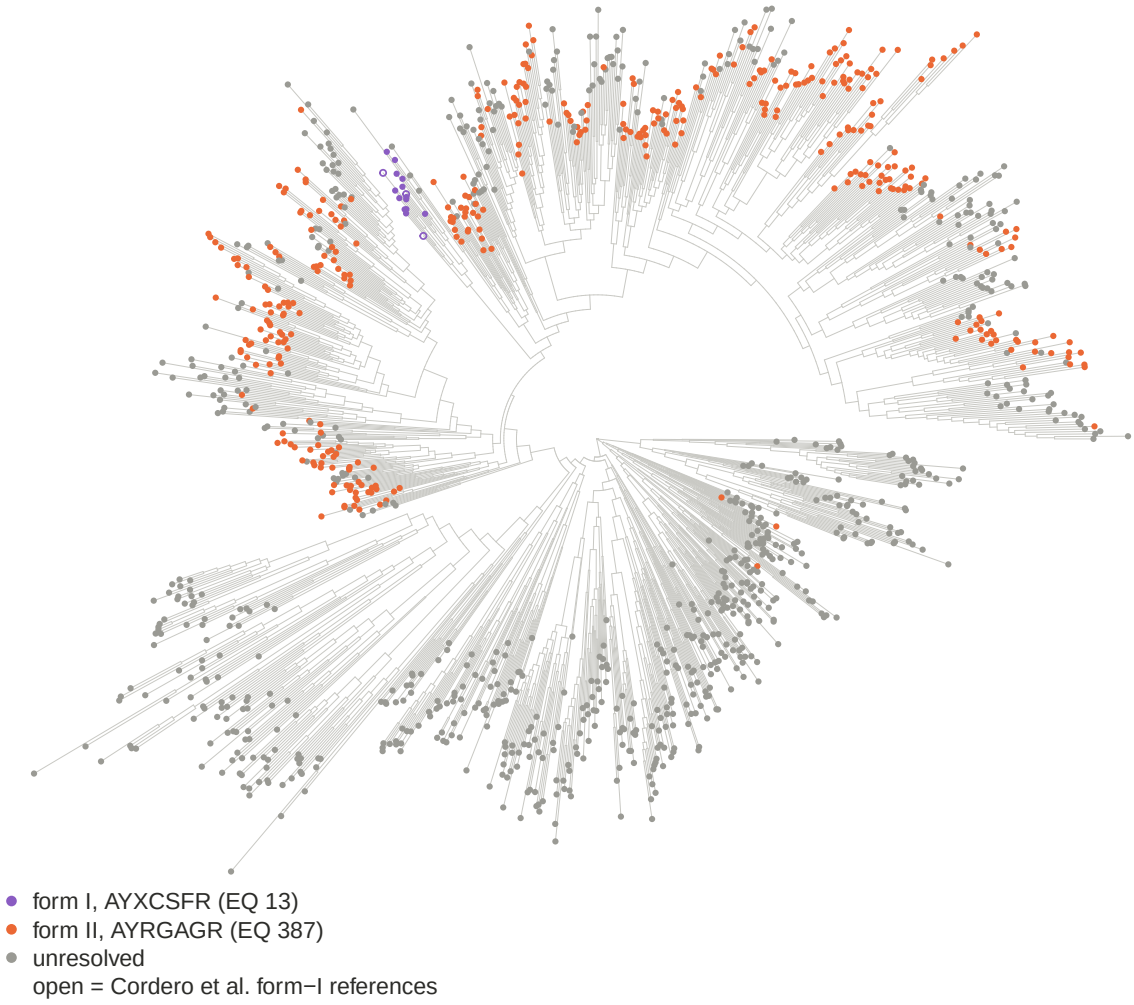


Figure S7: **CoxL large subunits.** Tree of Rub' al-Khali CoxL proteins of at least 600 residues (70% identity cluster representatives, filled) with the three form-I references of Cordero et al. [11] (open). Form I (AYXCSFR and specified variants) purple, form II (AYRGAGR) orange, unresolved grey.

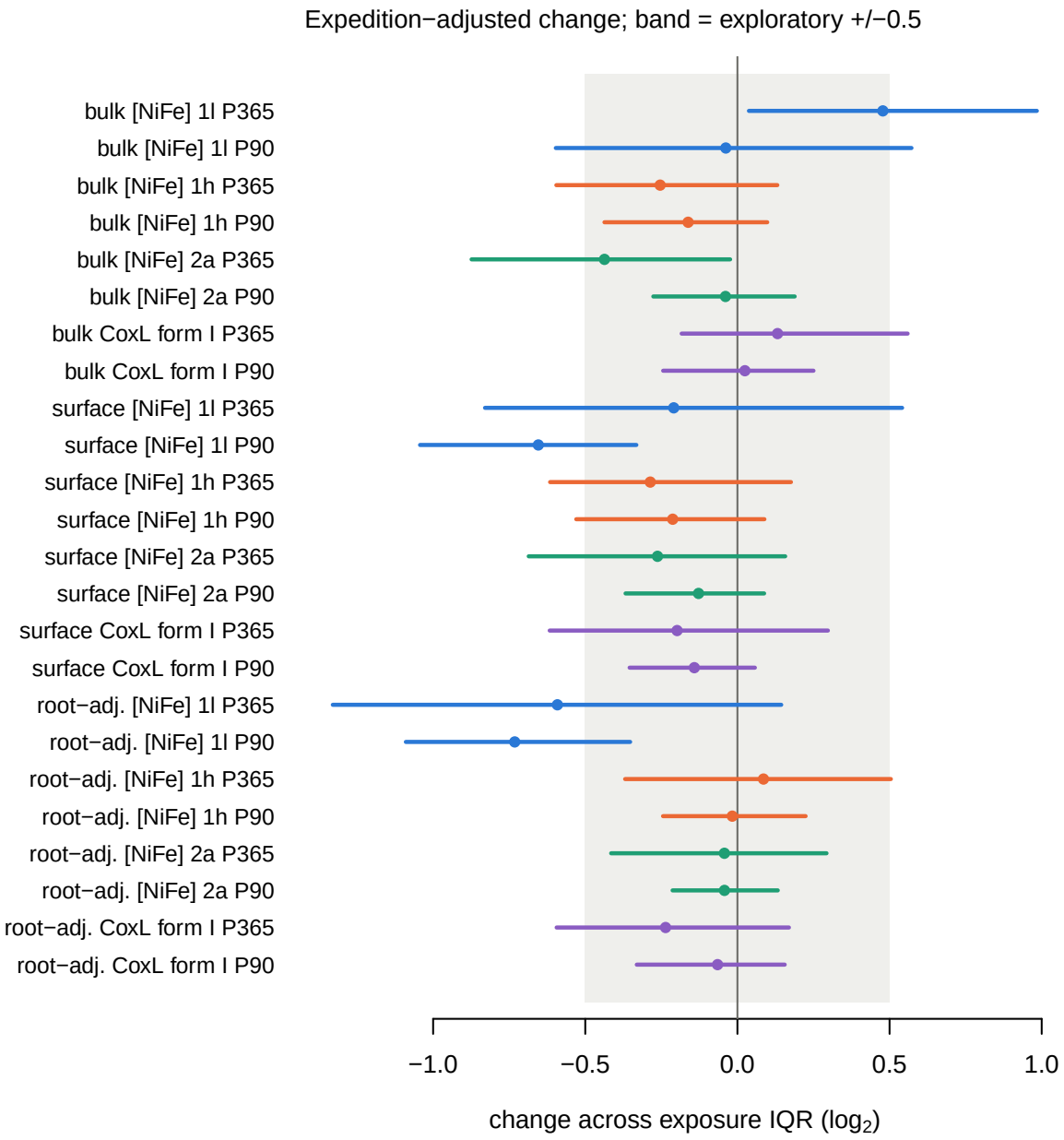


Figure S8: **Within-site change between expeditions.** Change in $\log_2(\text{GE}+0.01)$ across the within-site interquartile range of log(1+antecedent ERA5 precipitation) over 90 and 365 days, from mixed models with expedition as a fixed effect and site as a random intercept, with 95% intervals from resampling sites. The shaded band marks the exploratory ± 0.5 bound on the transformed scale.

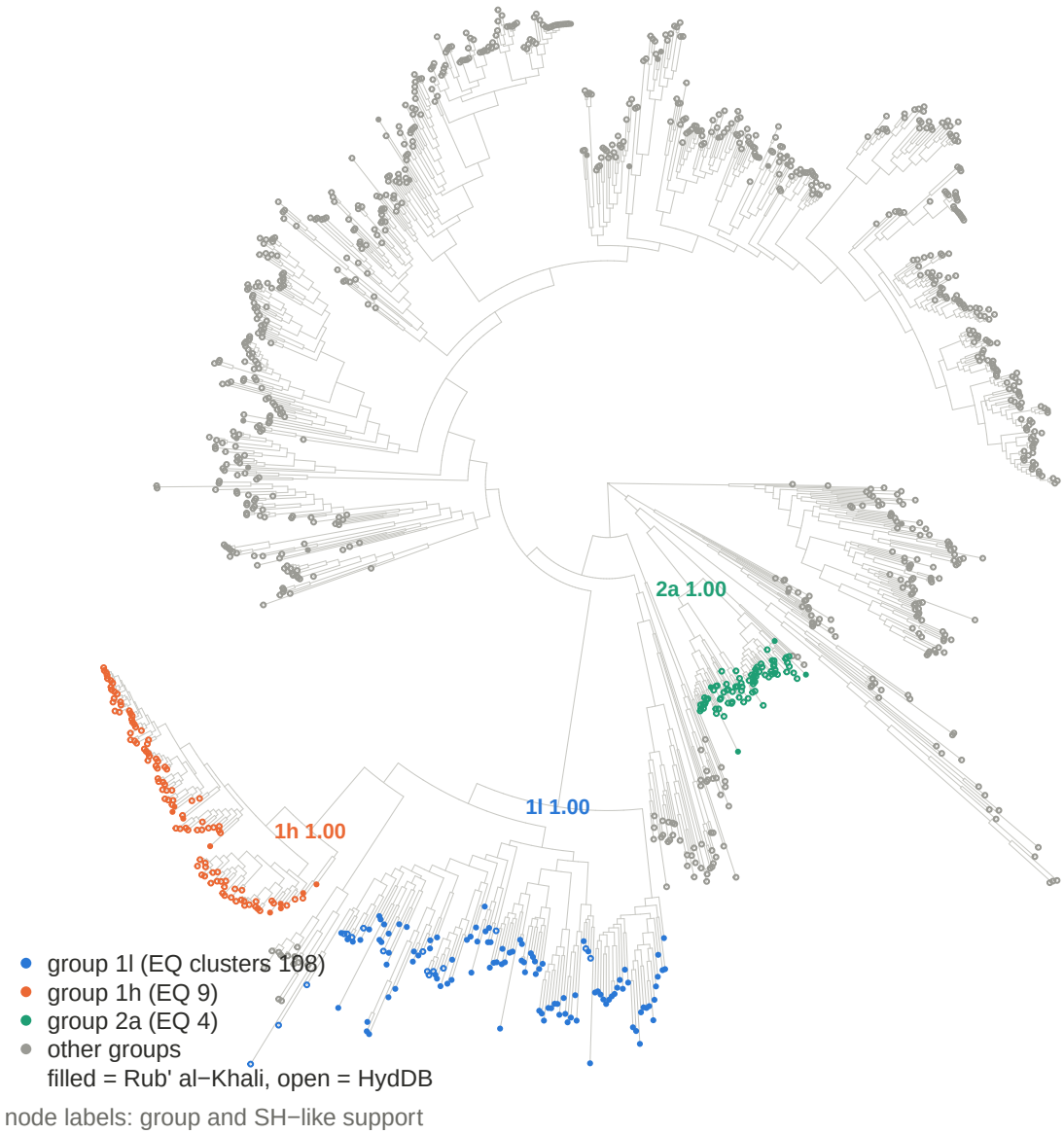


Figure S9: **[NiFe]-hydrogenase classification.** Large-subunit tree of catalogue proteins (70% identity cluster representatives, filled) and HydDB references (open), rooted on group 2. Colours identify groups 1l, 1h and 2a; node labels give SH-like support.

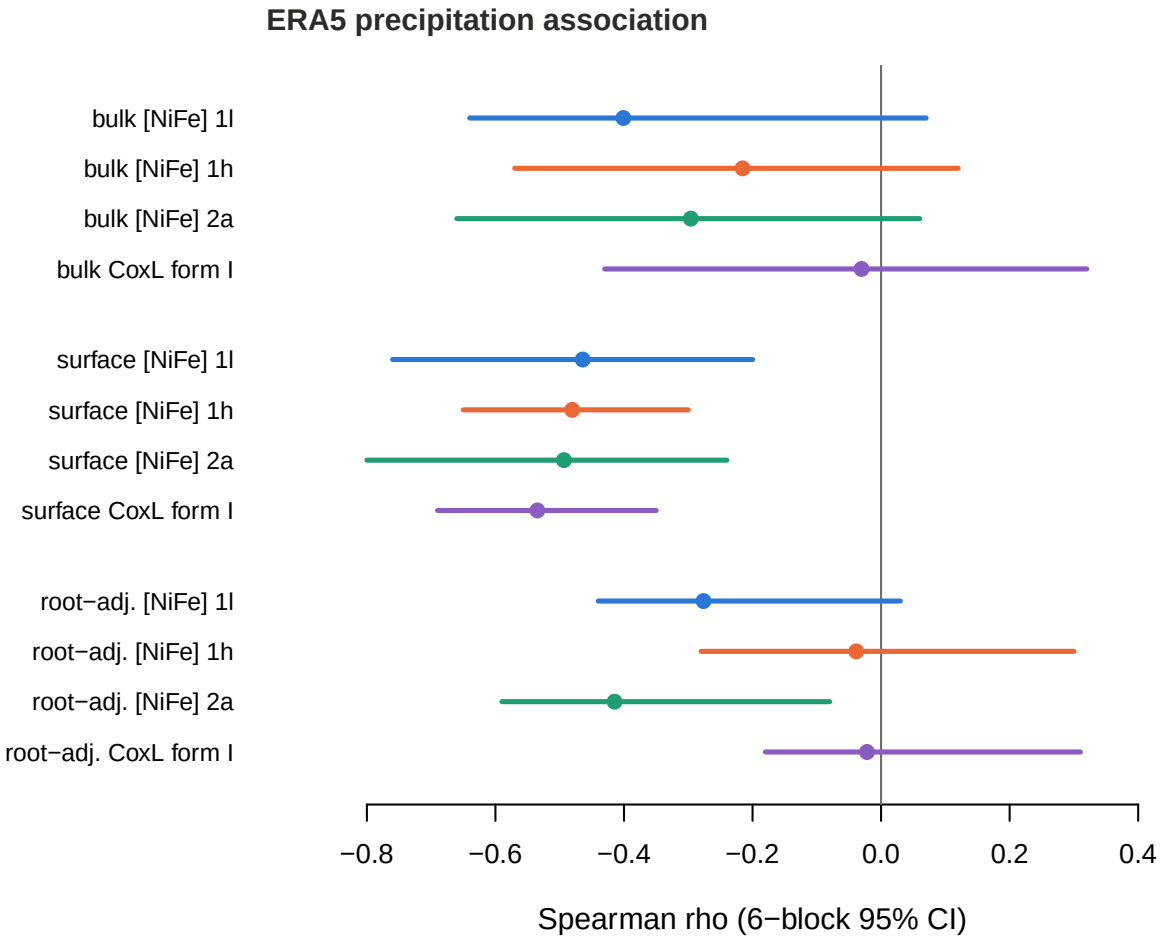


Figure S10: **ERA5 precipitation associations.** Spearman correlations of site medians with mean annual precipitation, with 95% intervals from resampling six spatial blocks. Product and block-count sensitivity are described in Supplementary Results.

Table S21: Surface-minus-subsurface sensitivity to the complete microTrait implementation. Values are percentage-point differences with 95% spatial-block bootstrap intervals, using identical paired membership (68 visits, 36 sites) and equal-site weighting. The earlier profile-screen estimates reproduce the canonical-taxonomy sensitivity results. These confidence intervals are separate from the INLA posterior intervals in Table S15.

Trait	Annotation	Positive MAGs	Mapped community	Original reads
BetL	Earlier profile screen	415	7.55 [4.72, 10.33]	3.30 [2.02, 4.56]
BetL	Full microTrait	341	6.91 [4.33, 9.45]	3.00 [1.84, 4.14]
AhpCF	Earlier profile screen	525	6.95 [3.34, 9.71]	3.23 [1.37, 4.83]
AhpCF	Full microTrait	522	6.93 [3.34, 9.68]	3.22 [1.37, 4.81]

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